Deep Learning
Final Project Handbook

Part A Shape and 3D force from GelSight tactile images (individual)
Part B Advanced tactile perception project and video (groups of 5–6)

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The project in one minute

- The project is marked out of **100**: **Part A is worth 60** and **Part B is worth 40**.
- **Part A (individual, due Sunday 13 December 2026):** train deep learning models on images from a *GelSight* vision-based tactile sensor to classify the *shape* pressed into it (10 classes) and estimate the *3D contact force* (f_x, f_y, f_z), including a required analysis of the tangential force. Submit a 3–6 page IEEE report, your code and your trained models. Your models are also run on a **hidden test set**.
- **Part B (group of 5–6, due Sunday 20 December 2026):** pitch an **advanced tactile perception** project that builds on a reading list of recent tactile research, in a video of **at most 5 minutes**. A working proof-of-concept model is mandatory, and every member presents. Groups are confirmed on **Monday 23 November 2026**.
- Up to **15 bonus marks** are available. Your final mark can never exceed 100.

If anything in this handbook is unclear, contact the module team (Section 1.3). All submissions go through Canvas.

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1 Welcome and overview

Your Deep Learning final project is worth **100 marks** in two parts. **Part A** is an individual technical coursework worth **60 marks**. **Part B** is a group project worth **40 marks**, done in groups of **5–6 students** and presented as a video. Up to **15 bonus marks** are available on top, but your final mark can never exceed 100. Both parts are about **tactile perception**: how a robot can understand touch from the images of a vision-based tactile sensor.

1.1 Part A: shape and 3D force from GelSight images (individual, 60 marks)

You build deep learning models for two tasks on images from a **GelSight** vision-based tactile sensor:

- **shape classification:** which of 10 shapes is pressed into the sensor (images from a GelSight *without* markers);
- **force estimation:** the 3D contact force f_x, f_y (tangential, or shear) and f_z (normal), in newtons, from a single image (images from a GelSight *with* a grid of markers). You must analyse the tangential force error separately for press and shear images.

You submit a 3–6 page IEEE-format report, your code and your trained models with a `predict.py` script. The module team runs your models on a **hidden test set** that you never see, using an automatic grading script, so your `predict.py` must follow a fixed interface exactly. Sections 3–5 cover Part A.

1.2 Part B: advanced tactile perception project (group, 40 marks)

In a group of 5–6, you choose an **advanced tactile perception** problem faced by a real stakeholder and pitch a deep learning solution in a video of **no more than 5 minutes**. "Advanced" means going beyond the single-image classification and regression of Part A: for example graph neural networks on tactile markers, cross-sensor generalization, visuo-tactile fusion, generative tactile data or hardness perception. Your project must build on at least one paper from the Part B reading list (Section 6.3). A working **proof-of-concept model** trained on tactile data is mandatory, and **every member presents** part of the video. Each member's Part B mark can be adjusted by **peer assessment**. Sections 6–8 cover Part B.

1.3 Who runs the project

The module is led by **Dr Dandan Zhang** (module leader, d.zhang17@imperial.ac.uk), supported by **Ms Lan Wei** (teaching assistant, l.wei24@imperial.ac.uk). Both review your Part A notebook, which is why you share it with the TA when you submit (Section 9.1). Bring questions to the module's teaching sessions or email the TA. If an answer changes anything in this handbook, the module team announces it to the whole class on Canvas.

1.4 What is new this year

Changes from the previous version of the project

- **Part A is new:** shape classification and 3D force estimation from GelSight tactile images.
- **Part A models are tested automatically** on a hidden test set through a standard `predict.py` interface (Section 5.3).
- **Part B is a group project** for 5–6 students, with roles, a group charter, meeting minutes,

contribution logs and peer assessment. It focuses on **advanced tactile perception**, building on a reading list of recent research, and must go beyond the Part A tasks.

- **Everything is submitted through Canvas.**

1.5 How to use this handbook

1. Read Sections 2 and 3 before you write any code. They say what is required, what is optional and where the marks are.
2. Use Sections 4 and 5 while you build and write up Part A.
3. Form your group in Weeks 1–2, and read Sections 6–8 together as a group.
4. Before you submit, work through Section 9 (how to submit), Section 13 (common pitfalls) and the checklists in Section 14.
5. Section 10 holds the full marking criteria. Section 15 answers common questions. Appendix A holds templates you can copy.

1.6 Materials you need

For	Files	Use them for
Part A data	`Tactile_DL_2027.zip`, from the Releases page of https://github.com/LannWei/Deep-Learning-CourseWork-2027	The GelSight dataset, its README (read it first) and `example_submission/`
Part A models	`example_submission/` inside the zip	A working `predict.py` with a small baseline CNN: the template for your model folder
Part A report	The IEEE conference template, from https://www.ieee.org/conferences/publishing/templates or "IEEE Conference Template" on Overleaf	Your 3–6 page report
Part B data	Public tactile datasets (Section 6.5)	Training your proof-of-concept model
Part B group work	Appendix A of this handbook	Charter, minutes, contribution log, storyboard, contributions and AI-use form
Both	This handbook and the briefing slides	Everything else

2 The project at a glance

Required work adds up to 100 marks: 60 for Part A and 40 for Part B. Optional work earns bonus marks, capped at 15, and your total can never go above 100. Figure 1 shows where the marks are.

Your final mark is

$$\text{Final} = \min\big(100,\ A + B \times PF + \mathit{Bonus}\big), \tag{1}$$

where A is your Part A mark (0–60), B is your group's Part B mark (0–40), PF is your peer factor (usually exactly 1, Section 10.3) and Bonus is at most 15. The product $B \times PF$ never exceeds 40.

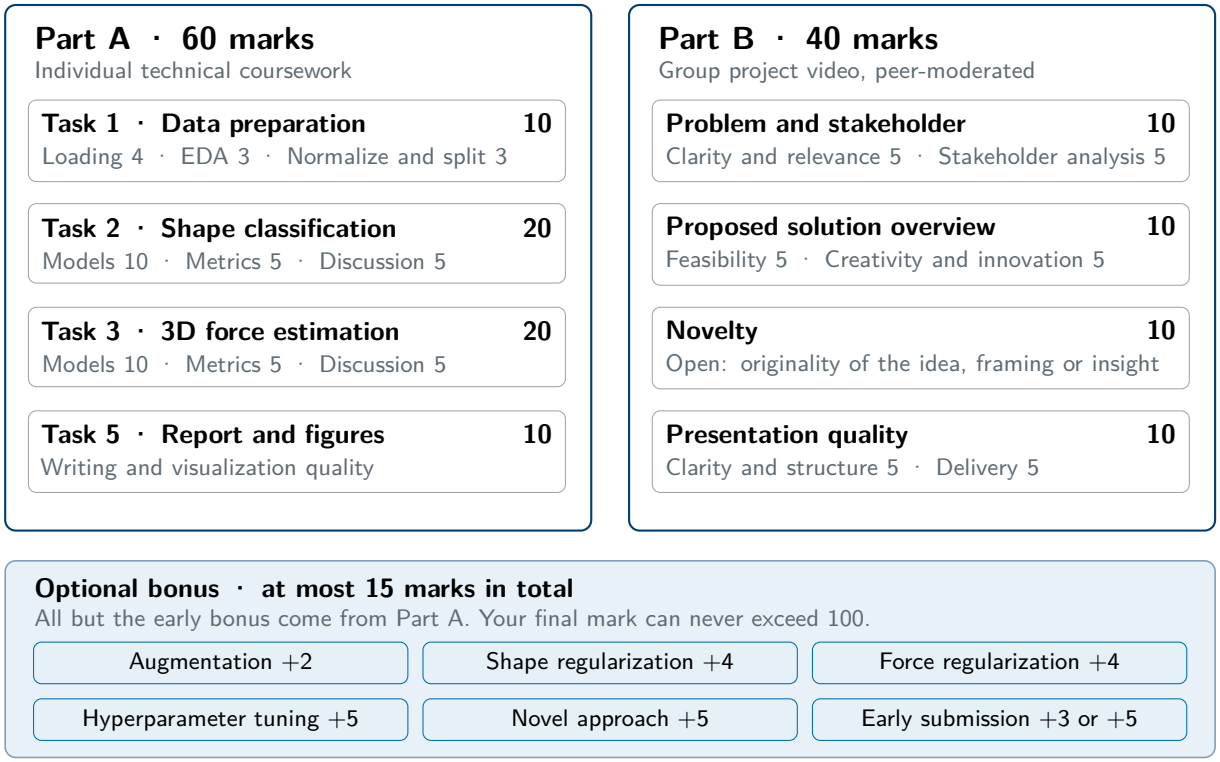


Figure 1: Mark structure. Part A is individual. Part B is a group mark that peer assessment can adjust for each member.

2.1 What you submit

Part	Deliverable	Who	Format and file name
A	One zip on Canvas	You	`DL_Coursework_YourName_CID.zip`, containing the three items below (Section 9.1)
A	Report	You	`DL_Coursework_YourName_CID.pdf`: IEEE conference format, double column, 3–6 pages
A	Code	You	Folder `code/`: your notebooks or scripts for preprocessing, training, evaluation and tuning
A	Model folder	You	Folder named with your CID, e.g. `01234567/`: `predict.py`, trained weights, helper modules, optional `requirements.txt`; under 200 MB
A	Notebook sharing	You	Share your Google Colab or Jupyter notebook with the TA, `l.wei24@imperial.ac.uk`
B	Video	Coordinator	`GroupXX_TactileDL_Video.mp4`, at most 5:00, with captions
B	Proof-of-concept code	Coordinator	`GroupXX_TactileDL_Code.zip`: notebook, README and model weights. Mandatory, not graded.
B	Contributions and AI-use form	Coordinator	`GroupXX_TactileDL_Contributions.pdf`, agreed by the whole group
B	Confirmation email	Coordinator	To the TA, copying every member

B	Peer assessment	Every student	Peer-assessment form on Canvas, after the Part B deadline

`YourName_CID` is your name and your 8-digit College ID (CID). For example, Jane Smith with CID 01234567 submits `DL_Coursework_JaneSmith_01234567.zip`. `XX` is your two-digit group number, e.g. `Group07`. Full instructions are in Section 9.

2.2 Key dates

Week 1 is the week commencing Monday 9 November 2026. All deadlines are 23:59 UK time (GMT).

Milestone	Date	What happens
Project released	Tuesday 10 November 2026 (Week 1)	Handbook, slides, dataset and example submission available
Group list confirmed	Monday 23 November 2026 (Week 3)	Every student is in a registered group of 5–6 on Canvas
Group charter and Part B direction	Sunday 29 November 2026 (Week 3)	Signed charter, and one paragraph on direction, dataset and stakeholder (feedback only, not marked)
Part B progress check	Week 4 (30 November – 4 December 2026)	10-minute meeting with the TA: proof-of-concept result and storyline
Part A deadline	**Sunday 13 December 2026, 23:59**	Zip on Canvas; notebook shared with the TA
Part B deadline	**Sunday 20 December 2026, 23:59**	Video, code zip and contributions form on Canvas; confirmation email
Peer assessment closes	Wednesday 23 December 2026, 23:59	Peer-assessment form on Canvas, from every student
Early bonus cut-offs	1 week / more than 2 weeks before a deadline	+3 / +5 bonus marks (Section 9.3)
Last day a late submission is accepted	7 days after a deadline	5 marks lost per day late

2.3 Required or optional?

- **Part A:** Tasks 1, 2, 3 and 5 are required, including the tangential force analysis in Task 3 and a `predict.py` that follows the interface. Task 4 (hyperparameter tuning), data augmentation, optimization and regularization, and a novel approach are optional and earn bonus marks only.
- **Part B:** the video, the proof-of-concept model and code, the contributions form and the peer assessment are all required. Collecting your own tactile data is optional.

Do the required work well first. The 100 required marks are worth far more than the 15 bonus marks.

3 Part A: the technical brief, task by task

Part A is individual work. It takes you from raw GelSight images to two trained models, one for shape classification and one for force estimation, and then to a `predict.py` script that the module team runs on a hidden test set. Tasks 1, 2, 3 and 5 are required; Task 4 is optional. Figure 2 shows the whole pipeline.

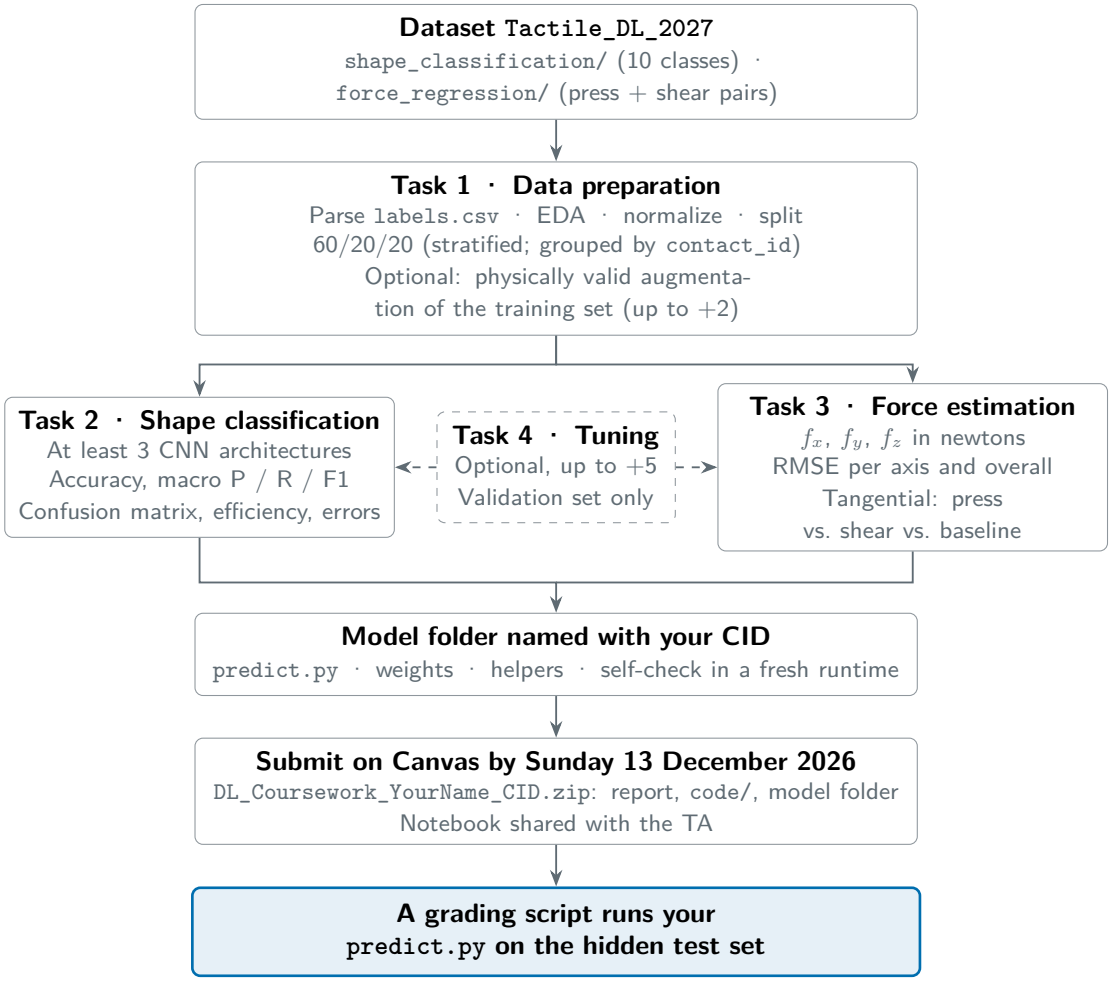


Figure 2: The Part A pipeline. Solid boxes are required; the dashed box is optional. Task 5, the report, documents every step. The last step is out of your hands, so your model folder must run exactly as the interface describes.

3.1 Background: vision-based tactile sensing

Tactile sensing plays a crucial role in robotics, prosthetics and human–robot interaction, where a robot must perceive the objects it touches and the forces it applies. Vision-based tactile sensors such as GelSight [1] convert contact into images: a soft elastomer gel with a reflective coating is lit from inside by coloured LEDs, and a camera behind it records the change in shading and colour when an object presses into the gel. The surface slope, and so the image, encodes the *geometry* of the contact. Some sensors print a grid of black markers on the gel. The markers move with the gel surface, and their displacement shows how the gel is stretched by *tangential (shear)* force.

Tactile images bring difficulties that natural images do not:

- **Small, subtle cues.** The contact often occupies a small part of an otherwise uniform image, and shallow contacts give low-contrast changes that can vanish when you downsample.

- **Position varies.** Contacts appear anywhere on the pad, including near the border and outside the marker grid.
- **Coupled forces.** Normal and tangential forces act together, and in many samples the tangential force is small compared with the normal force.
- **One image, no reference.** The model sees a single frame, not the undeformed gel, so it must infer marker displacement from that image alone.
- **Few labels.** Labels need robot-controlled data collection and a force/torque sensor.
- **Sensor-specific appearance.** The gel, illumination and camera of each sensor shape every image.

Shape classification identifies the geometry of the object in contact, which matters for object recognition when vision is occluded by the gripper or the object. **Force estimation** predicts the 3D contact force, which matters for grasp control, slip detection and safe physical interaction. The coursework data is a curated subset of the TacVerse benchmark [2]; sensor evaluation methods are discussed in TacEva [3]. Use these points to motivate your report's Introduction, in your own words and with citations.

3.2 The dataset

Download `Tactile_DL_2027.zip` (about 37 MB) from the Releases page of the coursework repository; *Code → Download ZIP* does not include it. **Read `README.md` inside the zip first**: it gives three ways to load the data in Colab, a minimal PyTorch `Dataset`, the split rules, the metrics and the submission interface. All images are 320×240 RGB JPEG.

```
Tactile_DL_2027/
|-- README.md
|-- example_submission/        # template for your model folder
|-- shape_classification/
|   |-- classes.csv            # class_name -> class_id
|   |-- labels.csv             # filename, class_name, class_id
|   `-- images/<class>/<class>_NNNN.jpg
`-- force_regression/
    |-- labels.csv             # filename, contact_id, is_shear, fx, fy, fz
    `-- images/contact_NNNN.jpg          # normal press
               contact_NNNN_shear.jpg    # same contact, after a lateral move
```

Shape classification. 2,000 images from a GelSight *without* markers, 200 in each of 10 classes (Figure 3). All images of a class come from the same physical indenter, pressed at different positions, orientations and depths, so many images within a class look very similar.

Force estimation. 2,000 images from a GelSight *with* markers, from 1,000 physical contacts. Each contact has two images that share a `contact_id`: a pure normal press (`is_shear` = 0) and the same press after a lateral move (`is_shear` = 1), as in Figure 4. The labels `fx`, `fy` are the tangential force components and `fz` is the normal force, all in newtons in the sensor frame. **Negative f_z means pressing into the sensor.** Table 4 gives the force statistics of the full source recording; your subset has a very similar distribution, which you should check yourself.

For normal presses, f_x and f_y are close to zero: large tangential forces only appear in the shear images. The three targets also have different scales, so standardize each one before training.

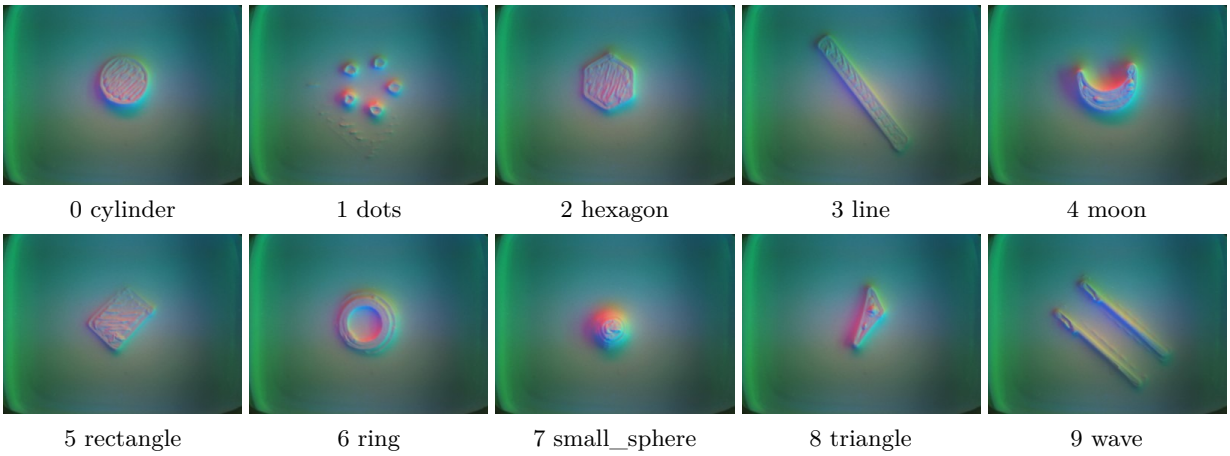
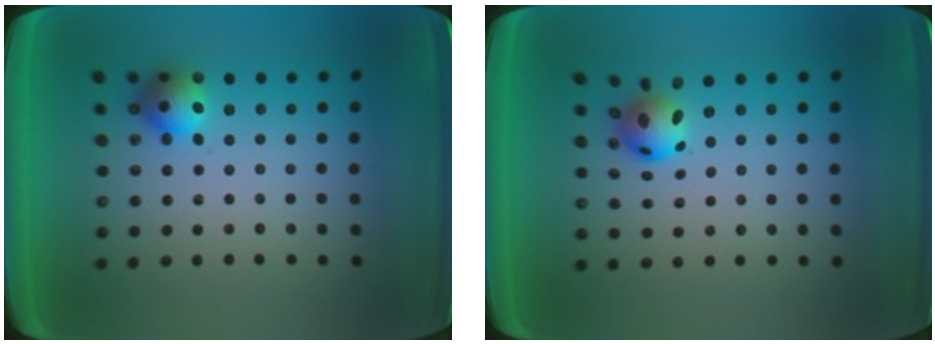


Figure 3: One example per shape class, labelled with `class_id` and `class_name` as in `classes.csv`.



press: $f_x = -0.09$, $f_y = 0.02$, $f_z = -1.99\,\mathrm{N}$ shear: $f_x = 1.40$, $f_y = -0.06$, $f_z = -1.85\,\mathrm{N}$

Figure 4: Contact 769: the normal press and the same contact after a lateral move. The markers around the contact are displaced by the shear.

3.3 Task 1: Data preparation and preprocessing

10 marks, plus up to 2 bonus marks for augmentation. The aim is to understand the dataset and get it into a clean, model-ready form.

Step	Required?	Marks	What to do
Data loading and structure understanding	Required	4	Load the images and parse both `labels.csv` files. Associate each image correctly with its shape label or its force label (f_x, f_y, f_z), including the press/shear pairs that share a `contact_id`.
Exploratory data analysis (EDA)	Required	3	Visualize sample images and analyse the class and force distributions. Identify key patterns, for example the near-zero tangential force in normal-press images.
Normalization and splitting	Required	3	Normalize pixel intensities and standardize the force targets. Split 60/20/20: **stratified by class** for shapes, **grouped by `contact_id`** for forces.
Data augmentation	Optional	+2 bonus	Implement augmentation (rotation, flipping, colour jitter, etc.) and justify that each choice is physically valid for tactile images.

Table 4: Force statistics of the full source recording (newtons).

	f_x	f_y	f_z
Press: mean / std	−0.07 / 0.06	−0.01 / 0.03	−0.73 / 0.47
Shear: mean / std	0.01 / 0.38	−0.03 / 0.42	−0.69 / 0.46
Overall min / max	−1.34 / 1.40	−1.37 / 1.28	−1.99 / 0.01

Good practice for Task 1

- **Use the file names in `labels.csv`** to find each image, then display a few images with their labels as a sanity check.
- **EDA that earns marks** shows sample images for both tasks including a press/shear pair, class counts, histograms of f_x, f_y and f_z *separately for press and shear*, and how f_z relates to the visible contact. Say what each plot tells you.
- **Split the force data by `contact_id`, not by image.** The press and shear images of one contact share the same press position and depth. If one is in training and the other in testing, your test score is over-optimistic. Use `GroupShuffleSplit` and check that no `contact_id` appears in two splits (code in Appendix A.1).
- **Stratify the shape split by class,** and report the counts per class and per split in a small table. Report the force statistics of each split too.
- **Fit normalization on the training set only**, then apply the same values to validation and test data. Keep the force mean and standard deviation: `predict.py` must return forces in newtons.
- **Fix your random seeds** so your split and results can be reproduced.
- **Ask what a random split measures.** Images within a shape class can be near-duplicates. Discuss whether your test split is a fair test of generalization.
- **Think about resolution.** 320×240 is not square. Resize, pad, crop or keep the aspect ratio, but justify the choice: small contacts may vanish if you downsample too much.

Augmentation must respect the physics. An augmentation must not change what the label means, or the label must change with it.

Augmentation	Shape classification	Force estimation
Flip	Usually valid: the class does not change.	**Labels must change**: a flip reverses the sign of one tangential component. Which one? The mapping between image axes and f_x, f_y is not given: work it out from the data and show your evidence.
Rotation	Usually valid. Watch the dark gel border and the empty corners that rotation introduces.	**Labels must change**: rotating the image rotates (f_x, f_y). It also rotates the marker grid, which never rotates in real data.
Translation or crop	Keep the whole contact in view; contacts can be near the edge.	Moves the contact relative to the marker grid and the gel border. Justify it.
Colour or brightness jitter	Reasonable: mimics illumination differences between sessions.	Shading encodes indentation depth and so f_z. Keep it mild and test the effect.
Blur or noise	Can erase shallow, low-contrast contacts.	Blurs the marker positions that carry the shear signal.

For the bonus, show an ablation (with and without each augmentation, on the validation set) and explain the result physically. Augment the training set only.

3.4 Task 2: Shape classification

20 marks, plus up to 4 bonus marks for optimization and regularization. The aim is to design and implement deep learning classifiers for the 10 shapes, and evaluate them with classification metrics.

Component	Required?	Marks	What is expected
Model design and implementation	Required	10	Implement and compare **at least three CNN architectures** tailored for 10-class shape classification. Experiment to balance accuracy and computational efficiency.
Evaluation metrics	Required	5	Accuracy, Precision, Recall, F1 score (macro-averaged) and a Confusion Matrix. Interpret the results clearly and compare models.
Discussion and analysis	Required	5	Insight into model performance, failure cases (e.g. confused shape pairs) and improvements.
Optimization and regularization	Optional	up to +4	For example dropout, weight decay, label smoothing, early stopping, learning-rate schedules, transfer-learning strategies, with evidence.

Choosing three architectures. Pick architectures that differ in a meaningful way, so the comparison teaches you something. Changing only the number of filters makes a weak comparison. One possible set:

Role	Example	What it tells you
Baseline	A small custom CNN with 3–4 convolution blocks, like the one in `example_submission/`	How hard the task is; a lower bound
Stronger or residual model	A deeper CNN with batch normalization or residual connections [4]	Whether more capacity helps on this data
Efficient or pretrained model	A pretrained MobileNet, EfficientNet or ResNet, fine-tuned	Whether ImageNet pretraining helps with gel images, and at what cost

You may add non-CNN models, such as a vision transformer, as extra comparisons; at least three of your models must be CNNs. Pretrained ImageNet models expect ImageNet normalization: say what you used.

Accuracy versus efficiency. For each model, report the number of parameters, the training time and the inference time per image (state the hardware) next to its accuracy and macro F1. A small table makes the trade-off obvious.

Make the comparison informative

A simple CNN already reaches high accuracy on a random stratified split, so three models with near-identical accuracy say little on their own. Add another axis of comparison, such as

efficiency, input resolution, training-set size, robustness, or a harder split, and explain what you learn from it. Report run-to-run variation (several seeds) where you can.

Evaluation, done properly.

- Choose models and settings on the **validation** set. Use your **test** split once, for the final numbers.
- Report Precision, Recall and F1 **macro-averaged**, and say so.
- Show the confusion matrix, ideally normalized per true class, and say which shapes are confused with which.

Error analysis that earns marks. Show a grid of misclassified images with true and predicted labels. Then look for patterns. Are the confused shapes alike in local geometry (edges, curvature, size)? Are errors concentrated in shallow presses, partial contacts or contacts near the border? Saliency maps or Grad-CAM [5] show whether a model looks at the contact or at the background. Propose a reason for each pattern and, where you can, test it.

3.5 Task 3: Force estimation (regression)

20 marks, plus up to 4 bonus marks for optimization and regularization. The aim is a regression model that predicts the 3D contact force from one image, accurately for both the normal and the tangential force.

Component	Required?	Marks	What is expected
Model design and implementation	Required	10	A regression model with three continuous outputs, f_x, f_y and f_z. Compare **at least two architectures**, and justify the output head and the loss.
Evaluation metrics	Required	5	RMSE per axis and overall; the **tangential error analysis** (press vs. shear, against a mean-prediction baseline); comparison across architectures.
Discussion and analysis	Required	5	Tangential vs. normal force errors, the role of marker displacement, and failure cases.
Optimization and regularization	Optional	up to +4	For example loss weighting between axes, robust losses, multi-task heads, schedules, label-aware augmentation, with evidence.

Designing the model. End the network with a linear layer of three units, trained on standardized targets with mean squared error, mean absolute error or Huber loss; explain your choice. Questions worth exploring: one head or separate heads for normal and tangential force? Should every axis count equally in the loss? Can the network still locate individual markers at your input resolution? Do features pretrained on natural images help with dots on gel? Can the press/shear pairs in the training data help, even though prediction uses one image?

Measuring the error. With $e_{ik} = \hat{f}_{ik} - f_{ik}$ the error of image i on axis k, in newtons, over the

N images of your test split, use exactly these definitions so results are comparable:

$$\mathrm{RMSE}_k = \sqrt{\frac{1}{N}\sum_{i=1}^{N} e_{ik}^2}, \qquad k \in \{x, y, z\}, \tag{2}$$

$$\mathrm{RMSE}_t = \sqrt{\frac{1}{N}\sum_{i=1}^{N} (e_{ix}^2 + e_{iy}^2)}, \tag{3}$$

$$\mathrm{RMSE}_{\mathrm{all}} = \sqrt{\frac{1}{3N}\sum_{i=1}^{N}\sum_{k} e_{ik}^2}. \tag{4}$$

RMSE_t is the error of the tangential force vector. Report every RMSE in newtons, not standardized units. MAE and R^2 are optional extras. Code is in Appendix A.1 and in the dataset README.

Required: the tangential force error analysis

A model that ignores shear can still look good on the overall RMSE, because half of the images have $f_x \approx f_y \approx 0$. So you must:

1. **Per-subset table.** Report RMSE_x, RMSE_y, RMSE_z and RMSE_t separately for normal-press images (`is_shear` $= 0$) and shear images (`is_shear` $= 1$).
2. **Trivial baseline.** In the same table, report the metrics for a baseline that always predicts the training-split mean force (the mean over all training images). This shows how much tangential force your model has actually learned.
3. **Prediction plots.** For shear images, plot predicted vs. true f_x and predicted vs. true f_y, with the line $y = x$.
4. **Discussion.** Compare the tangential error with the normal-force error and explain the difference. Which visual cues does your model need to estimate shear? Show and discuss your worst tangential predictions.

Optional: for shear images with a true tangential magnitude above 0.2 N, report the angular error between the predicted and true tangential directions.

	Press (`is_shear` $= 0$)				**Shear** (`is_shear` $= 1$)			
Model	f_x	f_y	f_z	t	f_x	f_y	f_z	t
Mean baseline	…	…	…	…	…	…	…	…
Model A	…	…	…	…	…	…	…	…
Model B	…	…	…	…	…	…	…	…

Suggested layout for the tangential table: RMSE in newtons; add the overall (press + shear) results too.

Discussion questions worth answering. Your model sees one image, not the press/shear pair, so where does it get a "reference" for marker displacement? Does the error depend on where the contact is relative to the marker grid? Is the f_z error the same in press and shear images, and why?

Must not. At prediction time your model may use only the image: not `is_shear`, `contact_id` or filename patterns. Hidden-test images are anonymous and not marked as press or shear.

3.6 Task 4: Hyperparameter tuning (optional)

Up to 5 bonus marks. The aim is to improve performance through **systematic** tuning and justify the values you chose with experimental evidence. Systematic means:

1. Define a search space: for example learning rate, batch size, weight decay, input size, augmentation strength, loss weights.
2. Choose a search method: grid search, random search or Bayesian optimization (for example Optuna).
3. Select on the validation set only.
4. Report the trials in a table or plot, and show how sensitive performance is to the most important settings.
5. Justify the final values from that evidence.

Trying a few values by hand and reporting only the best one is not systematic.

3.7 Task 5: Evaluation and reporting

10 marks, for report writing and visualization quality. The aim is to present your results clearly and in depth, with insight into performance, limitations and future directions.

- **Hidden test set.** The module team runs your models on unreleased images from the same sensors (Section 5.3). Aim for models that generalize, not ones tuned to your own test split.
- **Final report.** An IEEE-format paper covering Abstract, Introduction, Objectives, Methodology, Experiments, Evaluation, Conclusions and Future Work. Section 4 explains each part.

3.8 Optional: a novel approach

Up to 5 bonus marks for a creative, innovative approach that is well documented and justified. Ideas: hand-crafted marker features (for example marker displacements found by blob detection) combined with deep features, in the spirit of physics-informed learning; using the press/shear pairing as an extra training signal; or a model designed around the marker grid. The idea must be yours to explain, test and justify.

4 Part A: writing the report

Your Part A report is an IEEE conference paper: **double column, 3–6 pages**, built from the IEEE conference template. It earns Task 5's 10 marks directly. It is also where markers look for the evidence behind Tasks 1–4, so work that is not reported is hard to credit.

4.1 Format requirements

- **Template.** Use the official IEEE conference template (LaTeX or Word) from `https://www.ieee.org/conferences/publishing/templates`, or search for "IEEE Conference Template" on Overleaf. In LaTeX this is `IEEEtran.cls` in conference mode.
- **Tools.** The simplest route is Overleaf, or a local LaTeX installation, compiled with BibTeX.
- **Do not alter the layout.** Margins, column widths, line spacing and fonts are fixed by the template.
- **Title block.** Replace the placeholder title, name, department, organization and email. Delete any placeholder notes you do not use, such as the `\thanks{}` line.
- **Length.** 3–6 pages, counting everything up to the end of Conclusions and Future Work. References and the AI-use declaration may continue beyond page 6.

- **File name.** `DL_Coursework_YourName_CID.pdf`, inside your Part A zip.
- **AI use.** Add a short AI-use declaration after the references (Section 11.2).

A minimal skeleton in the template's style:

```
\documentclass[conference]{IEEEtran}
\usepackage{cite,amsmath,amssymb,graphicx,url}
\begin{document}
\title{Shape Classification and Contact Force Estimation from Vision-Based Tactile
    Images}
\author{\IEEEauthorblockN{Your Name}
\IEEEauthorblockA{\textit{Department} \\ \textit{Imperial College London} \\ email}}
\maketitle
\begin{abstract} ... \end{abstract}
\begin{IEEEkeywords} deep learning, tactile sensing, GelSight, ... \end{IEEEkeywords}
\section{Introduction}
\section{Objectives}
\section{Methodology}
\section{Experiments}
\section{Evaluation}
\section{Conclusions and Future Work}
\bibliographystyle{IEEEtran}
\bibliography{references}
\end{document}
```

4.2 Required sections

The brief requires these sections, in this order.

Section	What it should contain	Tips
Abstract	One paragraph: the problem, your approach, your key results with numbers, and the main conclusion	Give your best macro F1 and your force RMSEs, including the tangential error on shear images, with units written in words (newtons). No symbols, special characters, footnotes or maths.
Introduction	Background and motivation, the challenges (Section 3), related work with citations, and your contributions	Cite tactile sensing work, not only deep learning papers. End with a short list of what you did and found.
Objectives	What you set out to achieve, stated so it can be checked	For example: compare at least three CNNs for shape on accuracy and efficiency; beat the mean baseline on tangential force for shear images by a stated margin.
Methodology	Data pipeline (loading, EDA findings, normalization, split, any augmentation), model architectures, losses, training set-up	Use a table or diagram for the architectures. Give optimizer, learning rate, batch size, epochs, early stopping and hardware.
Experiments	What you compared and why: architectures, resolutions, hyperparameter search, ablations	Someone should be able to repeat your work from this section.

Evaluation	Results tables and figures, comparison across architectures, efficiency, the tangential analysis, error analysis, failure cases	Most marks for Tasks 2 and 3 are evidenced here. Interpret, don't just list numbers.
Conclusions and Future Work	Direct answers to your objectives, with the key numbers; the main limitations; concrete next steps	No new results here. "More data" is weak; "add contacts near the grid edge, where the tangential error doubles" is strong.
References	Every source you used, in IEEE numbered style	Generated by BibTeX.

A suggested page budget for a 6-page report

About 1 page for the Abstract, Introduction and Objectives; 1.25 for Methodology; 0.75 for Experiments; 2.25 for Evaluation; and 0.75 for Conclusions and Future Work. References go after that. This is a guide, not a rule, but Evaluation deserves the most space.

4.3 Figures and tables

Task 5 marks visualization quality as well as writing. Strong reports usually include:

- a split table: sizes, and the class or force distribution in each split;
- sample images for both tasks, including a press/shear pair, and the label distributions (class counts; force histograms for press and shear);
- training and validation curves for your final models;
- a shape architecture table: parameters, inference time, accuracy, macro precision, recall and F1;
- the confusion matrix of the best shape model, with misclassified examples;
- a force architecture table: per-axis and overall RMSE;
- the tangential table (press vs. shear, models vs. mean baseline) and predicted-versus-true f_x and f_y plots for shear images;
- your worst tangential predictions, shown and discussed;
- a hyperparameter results table, if you did Task 4.

Quality checklist for every figure.

- Readable at column width: text in the figure roughly as large as the caption text. Export as vector PDF where possible, high-resolution PNG otherwise.
- Axes labelled, with units (for example "Predicted f_x (N)").
- A caption that says what the reader should notice, not just what is plotted.
- Colour-blind-safe colours, and meaning that does not rely on colour alone.
- Model names identical in the text, tables and figures.
- In tables: the same number of decimal places throughout, the best result in bold, and a clear statement of whether numbers are validation or test results.

IEEE placement rules (from the template).

- Place figures and tables at the top or bottom of a column, not in the middle. Large ones may span both columns.

- Figure captions go **below** figures. Table titles go **above** tables.
- Write “Fig. 1”, even at the start of a sentence.
- Insert each figure or table after it is first cited in the text.

```
\begin{figure}[t]
\centering
\includegraphics[width=\columnwidth]{shear_fx_scatter.pdf}
\caption{Predicted versus true tangential force on the shear images of the
test split. Errors grow for contacts near the edge of the marker grid.}
\label{fig:shear}
\end{figure}
```

4.4 Writing style rules from the IEEE template

- **Title and abstract:** no symbols, special characters, footnotes or maths.
- **Abbreviations:** define each one the first time it appears in the main text, even if you defined it in the abstract, for example “root mean square error (RMSE)”.
- **Units:** use SI units. Put a zero before a decimal point: “0.25”, not “.25”.
- **Equations:** number them consecutively and define every symbol just before or after the equation. Refer to “(1)”, not “Eq. (1)”, except at the start of a sentence: “Equation (1) is . . . ”.
- **LaTeX cross-references:** use `\label` with `\ref` or `\eqref` rather than typed numbers. Put `\label` after `\caption`. Use `align` instead of `eqnarray`.
- **Common slips:** “data” is plural; there is no full stop after “et” in “et al.”; “i.e.” means “that is” and “e.g.” means “for example”; watch affect/effect and principal/principle.

4.5 References and citations

- Manage references with BibTeX in `references.bib` and cite with `\cite{key}`. IEEE style numbers citations in order, like [1].
- Good starting points: Goodfellow, Bengio and Courville’s *Deep Learning* [6]; LeCun, Bengio and Hinton in *Nature* [7]; He et al. on residual learning [4]; GelSight [1]; and the dataset paper, TacVerse [2]. Strong reports go further into the tactile sensing literature.
- Cite every architecture, pretrained model, dataset and library you use, and any code you adapted.
- Prefer peer-reviewed sources, and include DOIs where they exist.

5 Part A: code, model folder and `predict.py`

Your code must work for someone else. The module team reads your notebook, and a grading script runs your model folder on the hidden test set with no help from you. This section explains both.

5.1 Your code and notebook

- **Where:** a `code/` folder inside your Part A zip, holding the notebooks or scripts for data preprocessing, model training, evaluation and (if you did it) tuning. The code must reproduce the numbers in your report.

- **Comments:** explain each model's architecture and how each task is handled.
- **Runs cleanly:** use *Runtime* → *Restart session and run all* before you submit. It must run without errors.
- **Outputs kept:** save the notebook with its outputs (plots, metrics, training logs) so reviewers can see your intermediate results.
- **Shared with the TA:** share your Colab notebook with `l.wei24@imperial.ac.uk` (Section 9.1). The copy inside your zip is the version that is marked.

A structure that works well: (1) set-up: imports, library versions, random seeds; (2) data loading; (3) exploratory data analysis; (4) preprocessing and the 60/20/20 splits; (5) shape classification models; (6) force regression models, the mean baseline and the tangential analysis; (7) hyperparameter tuning (optional); (8) final evaluation on your test splits; (9) saving the best models; (10) building the model folder and running the self-check.

Any framework that is pre-installed in Google Colab is fine, for example PyTorch or TensorFlow/Keras. The example submission uses PyTorch.

5.2 The model folder

Your model folder is named with your CID (e.g. `01234567/`) and goes inside your Part A zip:

```
01234567/
|-- predict.py          # REQUIRED: inference for both tasks
|-- shape_model.pt      # your weights: any names and format
|-- force_model.pt
|-- my_models.py        # optional: helper modules that predict.py imports
`-- requirements.txt    # optional: extra pip packages, pinned
```

Start from `example_submission/` in the dataset zip. It is a complete, working folder with a small baseline CNN. Its `train_example.py` trains on all the data without a split: that is for demonstration only and is **not** an acceptable protocol for your coursework.

5.3 The `predict.py` interface

The grading script runs these commands **from inside your model folder**:

```
python predict.py --task shape --images <IMAGE_DIR> --out <OUT_CSV>
python predict.py --task force --images <IMAGE_DIR> --out <OUT_CSV>
```

- `<IMAGE_DIR>` is a **flat** folder of `.jpg` files: no class sub-folders and no labels. The files have anonymous names (`test_0001.jpg`, ...) and are 320×240 RGB, recorded with the same sensors and preprocessed exactly like the released data.
- `<OUT_CSV>` must contain **one row per image** with these columns:
  - `--task shape`: `filename, class_id` (an integer 0–9, as in `classes.csv`);
  - `--task force`: `filename, fx, fy, fz` (newtons, **original scale and sign**, not standardized).

  `filename` is the file name only, e.g. `test_0001.jpg`.
- The hidden results are scored with the same metrics you report: accuracy and macro F1 for shapes; per-axis, overall and tangential RMSE for forces. They are used as evidence of performance and generalization when marking Tasks 2 and 3. **A folder that does not follow the interface cannot be scored.**

5.4 Rules for the model folder

- `predict.py` contains, or imports from your own files, the model definition, all preprocessing and the code that loads your weights.
- Load weights relative to the script, e.g. `Path(__file__).parent / "shape_model.pt"`. No absolute paths such as `/content/drive/...`.
- Use a GPU if one is available, but the script **must also run on CPU**.
- **Nothing may be downloaded at prediction time**: no `pretrained=True`, no URLs. Ship every weight file in the folder.
- Each task must finish within **10 minutes** on a Colab GPU runtime.
- Keep the folder under **200 MB**.
- In `requirements.txt`, list only packages that are *not* already in Google Colab, one per line with a pinned version (e.g. `timm==1.0.9`). Never list `torch` or `torchvision`: the Colab versions are used.
- Work from the image alone: no `is_shear`, `contact_id` or filename patterns.

5.5 Saving your best models

Save the **best** version of each model, chosen on the validation set, not simply the last epoch. Save the force standardization constants with the weights, so `predict.py` can return newtons. In PyTorch:

```
import torch

best_val = float("inf")
for epoch in range(num_epochs):
    train_one_epoch(model, train_loader)
    val_loss = evaluate(model, val_loader)
    if val_loss < best_val:                  # keep the best epoch only
        best_val = val_loss
        torch.save({"state_dict": model.state_dict(),
                    "target_mean": mu.tolist(),   # plain lists, not NumPy arrays
                    "target_std": sd.tolist()},
                   "01234567/force_model.pt")
```

Save plain tensors and lists

Recent PyTorch versions load checkpoints with `weights_only=True`. Save tensors and plain Python lists, as the example does, not NumPy arrays or custom objects, or your checkpoint may fail to load in the grader. If you use Keras, save in the `.keras` format and load it the same way in `predict.py`.

5.6 The self-check

Before you submit, test your folder in a **fresh** Colab runtime, exactly as the grader will (dataset README, Section 4.3): copy a few released images into flat folders, `cd` into your model folder, install `requirements.txt`, run both commands and inspect both CSVs.

```
import shutil
from pathlib import Path
data = Path("/content/Tactile_DL_2027")
for task, pattern in [("shape", "shape_classification/images/*/*.jpg"),
```

```
                        ("force", "force_regression/images/*.jpg")]:
    dst = Path(f"/content/check_{task}"); dst.mkdir(exist_ok=True)
    for f in sorted(data.glob(pattern))[::99]:    # mixes press and shear images
        shutil.copy(f, dst)

%cd /content/01234567
!pip install -q -r requirements.txt
!python predict.py --task shape --images /content/check_shape --out /content/pred_shape.
    csv
!python predict.py --task force --images /content/check_force --out /content/pred_force.
    csv
```

Both commands must finish without errors, each CSV must have one row per image with the right columns, and the predictions should agree reasonably with the labels of the copied images. Then unzip your final submission zip in a new place and run the self-check again on the extracted folder.

5.7 Reproducibility

- **Seeds.** Fix the seeds of Python, NumPy and your framework (e.g. `torch.manual_seed(42)`) at the top of the notebook, and use a fixed `random_state` for the splits.
- **Versions.** Print the versions of your key libraries in the notebook output.
- **Paths.** Set the data location in one variable at the top of the notebook, so a reviewer only has to change one line.
- **Splits.** Save your split (e.g. the file names in each split) or the code that recreates it.

6 Part B: the advanced tactile perception brief

In **Part B**, your group of 5–6 pitches a deep learning solution to an **advanced tactile perception** problem faced by a real stakeholder, in a video of **at most 5 minutes**. Your idea must build on at least one paper from the reading list (Section 6.3), and you must back it with a **proof-of-concept model** trained on tactile data. This section explains what "advanced" means, the research, the directions you can choose and the minimum technical work. Section 7 covers the video and Section 8 covers group working.

Proposed solution versus proof of concept

Your **proposed solution** is the full system you pitch to the stakeholder, for example a tactile fingertip that warns a surgeon about stiff tissue in real time. It can be ambitious. Your **proof of concept** is the small model you actually build to show the core idea works, for example a model that classifies hard versus soft contacts from tactile images and shows which image regions drove the decision. A full implementation of the proposed solution is *not* required; a working proof of concept *is*.

6.1 What "advanced" means

Part A asks for single-image classification and regression on one sensor. Part B must go further, in at least one of these ways:

- **a different kind of model,** such as a graph neural network on tactile markers, a generative model or a self-supervised encoder;
- **a harder setting,** such as a new sensor (cross-sensor generalization), very few labels, or

several frames over time (slip);

- **another modality,** such as vision and touch together, or translation between them;
- **a new tactile property or task,** such as hardness, material or texture, with an explanation the stakeholder can trust.

Part B may not use the Part A coursework dataset: Part A is individual work, and a group project on the same data and tasks would blur that line.

6.2 Tactile perception in brief

Tactile perception means inferring what is happening at a contact from touch sensors: what is being touched, where, how hard, and whether it is slipping. Many modern tactile sensors are **vision-based**: a small camera films a soft skin or gel as it deforms. Examples include GelSight and DIGIT, and the TacTip family of biomimetic sensors, whose skin carries pin-like markers that move when the skin is pressed. Every reading is therefore an image, and convolutional neural networks (CNNs) apply directly. Other sensors give arrays of pressure values (taxels) or time series.

Touch matters wherever vision alone is not enough: dexterous robot manipulation, grasp and slip detection, prosthetic hands, surgical palpation (finding stiff tissue by touch), quality inspection and assistive robotics. Tactile data is hard to learn from because:

- **Labelled datasets are small,** and labels need careful experiments.
- **Sensors vary:** skin wear, lighting and manufacturing differences change the images, and models trained on one sensor often fail on another [2].
- **Frames are correlated:** neighbouring frames from the same press or slide are almost identical, so a random split by frame leaks test data into training.
- **Contact is dynamic:** slip and texture often need several frames, not one.
- **Errors have consequences:** when a robot acts on a prediction, mistakes can damage objects or hurt people.

6.3 The Part B reading list

Table 12 lists the most relevant papers for each direction (Section 6.4). **Read at least two of them as a group before you choose a direction**. Your idea must build on at least one paper from this list, and your video must say which.

Table 12: The Part B reading list: the most relevant papers for each direction.

Paper	What it does	Read it for
TactEx [8]	An explainable multimodal framework for human-like touch and hardness estimation.	Direction A: hardness by touch, with explanations a user can trust.
ViTacTip [9]	A sensor with a transparent skin and biomimetic markers, so the camera sees the object as well as the contact.	Directions A and F: texture and grating cues; contact and force sensing.
Tac-VGNN [10]	Treats the markers of a TacTip sensor as the nodes of a graph and uses a graph neural network, with Voronoi-cell features linked to contact depth, to estimate contact pose.	Direction B: turning a marker image into a graph; graph model versus CNN.

Paper	What it does	Read it for
TacVerse [2]	A benchmark of tactile images from seven vision-based sensors for shape classification, grating classification and force regression, with within-sensor, cross-sensor and few-shot protocols.	Direction C: evaluation across sensors.
SARL [11]	Spatially-aware self-supervised representation learning for visuo-tactile perception.	Directions C and D: learning from unlabelled data.
Sparsh [12]	Self-supervised touch representations for vision-based tactile sensors.	Direction C: pretrained tactile encoders.
Multi-modality sensor with GAN interpretation [13]	Two Pix2Pix-based GANs translate visuo-tactile images to imitate a touch-only or a vision-only sensor.	Direction D: cross-modal translation.
MultiDiffSense [14]	A ControlNet diffusion model that generates images for three sensor designs from an object depth map and a text prompt giving the contact pose.	Direction E: synthetic tactile data.
The Feeling of Success [15]	Predicts grasp outcomes from GelSight images on the gripper fingers, with over 9,000 recorded grasps.	Direction F: grasp-outcome and slip detection.
MagicTac [16], **CrystalTac** [17]	Vision-based tactile sensors 3D-printed in one piece: an internal multi-layer grid (MagicTac) and a family of multi-material designs (CrystalTac).	Any direction: low-cost sensors if you collect your own data.

How to use the research in your video

Say which paper inspired your direction and what your group adds: a new stakeholder, a different model, a harder setting or a new application. Show the reference on screen and in your README. You are *not* expected to build these sensors; public datasets are fine for the proof of concept.

6.4 Choose one direction

Each group chooses **one** direction and tells the module team by **Sunday 29 November 2026**, in one paragraph naming the direction, the paper(s) it builds on, the dataset and the stakeholder. This is for feedback only and is not marked. Direction G must be approved at this point.

Direction	Model or setting	Read first	Suggested data	Example stakeholders
A. Material, texture or hardness perception, with explanations	Classification plus explainability	TactEx; ViTacTip	Touch and Go; ObjectFolder Real; TacVerse grating data	Quality inspection, prosthetic-hand users, fruit picking
B. Graph-based learning on tactile markers	Graph neural networks	Tac-VGNN	Marker-based images (TacVerse, TacTip-family sensors)	Robotics researchers, surface inspection
C. Cross-sensor generalization and self-supervised pre-training	Transfer, few-shot or self-supervised learning	TacVerse; SARL; Sparsh	TacVerse (sensors other than GelSight); Touch-Slide	Warehouses with mixed grippers, sensor makers
D. Visuo-tactile fusion and cross-modal translation	Fusion or GAN translation	Multi-modality sensor (GANs); SARL	Touch and Go; ObjectFolder Real	Robots working in poor lighting
E. Generative tactile data	Diffusion or GAN data generation	MultiDiffSense	A small real dataset plus generated images	Start-ups with little labelled data
F. Slip and grasp-outcome detection	Multi-frame (temporal) classification	The Feeling of Success; ViTacTip	Feeling of Success; YCB-Slide	Warehouse operators, assistive robotics
G. Your own idea	Any advanced setting (Section 6.1)	Any paper in Table 12	Agreed by 29 November	Your choice, for example medical robotics

6.5 Datasets you may use

These datasets are large. A documented subset (for example 5–10 classes and a few thousand frames) is fine. Check each licence, cite the dataset, and never include raw data in your submission. The Part A coursework dataset may not be used in Part B.

Dataset	Sensor	What it contains	Notes
TacVerse [2]	Seven vision-based tactile sensors	Images for shape classification, grating classification and force regression	Download via `https://github.com/LannWei/Tactile_Database`. Do not use the GelSight subsets in Part B, and never use any TacVerse data in Part A
Touch and Go [18]	GelSight, plus RGB video	Human-collected touches of objects in natural scenes; material labels on touch-onset frames	CC BY 4.0 licence
ObjectFolder Real [19]	Gel-based tactile video	100 household objects; tactile, visual and audio data at 30–50 contact points per object	Downloads come in batches of 10 objects
Feeling of Success [15]	GelSight on each gripper finger, plus camera	Over 9,000 robot grasp trials and their outcomes	HDF5 files

YCB-Slide [20]	DIGIT	Sliding touches over YCB objects, with sensor poses and object meshes	Best for object recognition; localization is advanced
Touch-Slide	DIGIT	About 180k unlabelled frames from 9 objects	Self-supervised pre-training only; CC BY-NC 4.0

Pretrained tactile encoders such as Sparsh [12] are allowed as an advanced option, as long as you train and evaluate the model yourselves.

6.6 The proof-of-concept model: minimum requirements

Every group must do all of the following. Nothing beyond this list is needed for a good mark.

1. **One advanced task, one dataset** (Section 6.1). A documented subset is fine.
2. **A leakage-free split.** Split into training, validation and test sets (60/20/20 or 70/15/15) **by object, session, trial or sensor, never by single frame**. Normalize with training-set statistics only.
3. **A model you trained yourselves,** such as a small CNN or graph network. Fine-tuning a pretrained network such as ResNet-18 [4] is also fine.
4. **A reference point.** Compare with a trivial baseline (majority class, or predicting the mean) or a simpler model, so your number means something. For a cross-sensor project, the within-sensor score is a natural reference.
5. **A test-set result.** Classification: accuracy, macro F1 and a confusion matrix. Regression: RMSE, MAE and a predicted-versus-true plot.
6. **At least one failure case** that you can show and explain.
7. **A short demo clip** of the model predicting on test examples, as a screen recording or with a live sensor.

The code is submitted as supplementary material (Section 9.2). It is **not graded**, but it is **mandatory**: markers use it to check that the proof of concept shown in your video is real.

Computing. Google Colab with a free GPU is enough. Resize images (for example to 128×128 or 224×224), work on a subset, and save checkpoints to Google Drive so a disconnect does not cost you a training run. TensorFlow/Keras and PyTorch are both fine for Part B.

6.7 Collecting your own tactile data (optional)

Data your group collects can make a proof of concept more convincing, but it is optional: the Creativity and innovation criterion rewards novelty in *methods*, and novelty in the dataset is not the key.

- Agree it with the module team first, by the direction deadline (Sunday 29 November 2026).
- Use robots or sensors only after a safety induction and with the TA's agreement.
- Touching objects normally needs no ethical approval, but anything involving people (touching skin, filming people, user studies) needs College ethical approval **before** you start.
- Keep the data in College storage (OneDrive or Teams), not personal accounts, and document how you collected and labelled it.

7 Part B: making the video

Your video has **at most 5 minutes** to explain an advanced tactile perception problem, your stakeholder, your proposed deep learning solution with its proof of concept, and what is new about it, with every member presenting. Five minutes for 5–6 speakers is tight, so plan the video as one piece from the start.

7.1 Audience

Present to an imagined panel of three people: a **robotics engineer**, a **potential user** (such as a surgeon, a warehouse manager or a prosthetic-hand user) and a **funder**. Each should leave knowing what problem you solve, for whom, how well your proof of concept works, what is new, and what could go wrong.

7.2 Requirements

Requirement	Detail
Length	**Hard limit 5:00.** Markers stop watching at 5:00, so anything after that earns nothing. Aim for 4:30–4:50.
Presenters	Every member is seen or heard presenting at least one segment, roughly 40–50 seconds each. Name each speaker on screen when they first appear.
Own voices	Members present in their own voices. AI voice clones, avatars or text-to-speech may not replace a presenter, unless agreed with the module team as a reasonable adjustment.
Proof of concept	At least one segment shows your model's test result against its reference point, and a short demo on real test data.
Research link	Name the paper(s) from Table 12 that your idea builds on, with the reference on screen.
Opening card	Group number, project title and member names, for about 5 seconds.
Closing card	Who did what, datasets used, and any third-party images, footage or music with their licences, for about 10 seconds. Both cards count towards the 5:00.
Captions	Required: burned into the video or supplied as a matching `.srt` file. Auto-generated captions are fine once you have corrected them.
File format	MP4 (H.264 video, AAC audio), 1920×1080 preferred, 1280×720 minimum. Keep the file under 1 GB; if Canvas will not accept it, see Section 9.2.

7.3 Addressing each marking criterion

The video is marked on four criteria of 10 marks each (full table in Section 10).

1. Problem and stakeholder (10 marks)

Markers look for a clearly defined problem whose relevance to the stakeholder is well argued (clarity and relevance, 5 marks), and a stakeholder analysis covering their needs, objectives and potential impact (5 marks).

- **Name a specific stakeholder,** for example colorectal surgeons doing minimally invasive surgery, or operators of a parcel-sorting robot, not "society".

- **State the problem in one sentence** the stakeholder would agree with, and say why touch helps where vision alone does not.
- **Answer three questions:** What does the stakeholder need? What are they trying to achieve? What changes for them if the problem is solved? Say what "good enough" means for them: accuracy, speed, safety or cost.

2. Proposed solution overview (10 marks)

Markers look at feasibility analysis (5 marks) and at creativity and innovation (5 marks).

- **Feasibility:** your methods, models, datasets and metrics must be realistic and suit the problem. Explain why a deep learning approach fits, which data you would use and how it would be labelled, how you would measure success, and what could go wrong (Section 7.6).
- **Show the proof of concept:** your test result against the reference point, the demo, and one failure case. Say honestly what it does and does not show.
- **Creativity and innovation:** show novelty in the **methods**: the architecture, the learning strategy, the combination of sensing, or how you evaluate. Novelty in the dataset is not the key, so a new dataset alone earns little here.

3. Novelty (10 marks)

This criterion is open. Markers look for originality in the idea as a whole: a framing, insight or combination that a stakeholder or an expert would not have thought of. Make the case explicitly.

- **Position your idea against the papers you build on.** What do they already do, and what does your group add?
- **Compare with the obvious approach.** Why is this not just a standard model applied to a standard problem?
- **Back the claim with evidence** where you can, for example a proof-of-concept result that the obvious approach does not reach, or a failure case it does not handle.

4. Presentation quality (10 marks)

Markers look at clarity and structure, including keeping within 5 minutes (5 marks), and at engagement and delivery: engaging, confident, professional, clear and concise (5 marks). For a group, delivery also means every member presents and the video works as one coherent piece rather than six separate talks.

7.4 A suggested 5-minute running order

Figure 5 and the table below are a starting point, not a rule. The plan ends at 4:45, leaving 15 seconds of margin. In a group of six, each content segment has one speaker; in a group of five, one member takes two short segments.

Time	Speaker	Content	Criterion
0:00–0:05	–	Opening card: group number, title, members	–
0:05–0:45	Member 1	Hook and problem: what tactile perception is, your problem, why touch	Problem and stakeholder

0:45–1:25	Member 2	Stakeholder: who they are, what they need, what success means for them	Problem and stakeholder
1:25–2:15	Member 3	Proposed solution: sensor and data, model, metrics, feasibility and key risks	Proposed solution
2:15–3:05	Member 4	Proof of concept: result against the reference point, demo, one failure case	Proposed solution
3:05–3:55	Member 5	What is new: the papers you build on, what you add, compared with the obvious approach	Novelty
3:55–4:35	Member 6	Impact and summary: what you propose and why the stakeholder should adopt it	Presentation quality
4:35–4:45	–	Closing card: who did what, datasets, credits	Presentation quality
4:45–5:00	–	Margin: never plan to the last second	–

7.5 Making a good video in five minutes

- **Storyboard first.** One row per shot: what is said, what is shown, who speaks. Agree it as a group before anyone records. A template is in Appendix A.5.
- **Script and time the narration.** About 130–150 words a minute is comfortable, so 4:30 of speech is roughly 600–675 words: about 100–110 words per speaker in a group of six.
- **Show, don't read.** One idea per slide, text of 24 pt or larger, figures instead of bullet points, never a full-page table.
- **Show the tactile images.** Most viewers have never seen one. A few seconds of raw sensor footage explains the problem better than a paragraph.
- **Make it one video, not six talks.** Use one slide template, one colour scheme and planned hand-overs ("Now Priya will show how our model performed").
- **Sound matters more than picture.** Record in a quiet room with a headset or a phone microphone close to your mouth. Keep music quiet under speech, or leave it out.
- **Watch it together before submitting.** Check timing, sound levels, captions and spelling.

Any tool is fine: PowerPoint's built-in recording, Teams or Zoom recordings, OBS Studio, Clipchamp, DaVinci Resolve or Kdenlive.

7.6 Risks and responsible use

A feasible solution names what could go wrong for its stakeholder and what you would do about it. Pick the one or two risks that matter most for your stakeholder and go deep, rather than listing all of them. A mitigation should be something a team could actually do, not a general intention such as "be careful with data".

Risk area	Tactile example	Possible mitigation
Safety	A missed slip makes a robot drop a heavy or fragile object near a person	Confidence thresholds, a safe fallback grip force, a human in the loop when confidence is low

Robustness	Accuracy drops when the sensor skin wears, the lighting changes or the sensor is replaced	Test on another session or sensor, a recalibration routine, clearly stated operating limits
Bias and fairness	A prosthetic-hand model trained on one person's grasps works worse for others	Data from diverse users, per-user calibration, results reported per user group
Privacy	Visuo-tactile datasets film homes and bystanders	Blur faces, strip location data, collect only with consent
Transparency	A user cannot tell why the system judged a grasp safe	Show confidence, explainability maps, documented limits
Synthetic data	Generated tactile images contain artefacts the real sensor never shows	Never test on generated images; report real-only test results

7.7 Rights, consent and accessibility

- Use only images, footage and music you have the right to use, and credit them in the closing card. Figures from the papers in Table 12 must be cited on screen.
- Get consent from anyone filmed who is not in your group.
- Keep personal data off screen, such as email inboxes, student ID numbers or other people's faces in the background.
- Use colour-blind-safe palettes and strong contrast; never rely on colour alone to carry meaning.
- The module team may ask to show strong videos to future cohorts. This needs every member's consent, and saying no has no effect on your mark.

8 Part B: working as a group

Each Part B group has 5–6 members. Every member leads one role, contributes to the technical work and presents part of the video. Part A remains individual work: do not share Part A code, results or reports within your group.

8.1 Forming your group

- Form your own group of 5–6 in Weeks 1–2 and register it on Canvas. **The group list is confirmed on Monday 23 November 2026.** The module team gives each group a number (Group01, Group02, . . .) used in all Part B file names.
- Anyone without a group by 23 November is placed by the module team. Groups smaller than five may be merged or topped up.
- Groups are fixed from 23 November, except through the problem process below.

8.2 Roles

Role	Leads on	Typical tasks

Coordinator and stakeholder lead	Planning, submission and the stakeholder	Runs meetings, keeps minutes and the action list, tracks deadlines, leads the stakeholder analysis, makes the final upload and sends the confirmation email
Research and novelty lead	Research link and what is new	Leads the reading of the papers in Table 12, positions the idea against them, builds the novelty argument and the reference list
Data lead	Dataset and pipeline	Downloads and subsets the data, sample plots, leakage-free split, data loaders, any own data collection
Modelling lead	Proof-of-concept model	Reference point, model training, checkpoints, the clean notebook
Evaluation and demo lead	Results	Metrics, plots, failure cases, the demo clip, risks and operating limits
Video producer	The video	Storyboard, script, recording plan, editing, captions, media licences

In a group of five, combine two roles, for example modelling with evaluation. Leading a role does not mean doing it alone: the lead makes sure it gets done.

Three rules for every group

1. **Everyone contributes to the technical work** (data, code, experiments or analysis). Nobody only edits the video.
2. **Everyone presents** part of the video, in their own voice.
3. **Everyone reviews** the final video, code and contributions form before submission. Submitting means the whole group agrees with what is submitted.

8.3 Group charter (by Sunday 29 November 2026)

Agree a one-page charter, have every member sign it, and upload it to Canvas by **Sunday 29 November 2026**, together with your one-paragraph direction (Section 6.4). The charter is not marked, but it is required, and the module team uses it if problems arise. It records members and roles, how you communicate and how fast you reply, your weekly meeting time, how you make decisions, internal deadlines, and what happens when someone misses one. A template is in Appendix A.2.

8.4 Meetings and records

- Meet at least once a week, in person or online, for 30–60 minutes.
- Keep minutes in a shared document: date, attendance, decisions, and actions with an owner and a due date (Appendix A.3).
- Share your work in one place: a shared drive or Teams channel for files, and GitHub or a shared Colab notebook for code.
- Each member keeps a short contribution log: date, task, hours and a link to the evidence (Appendix A.4). It feeds the contributions form and protects you if peer marks are disputed.

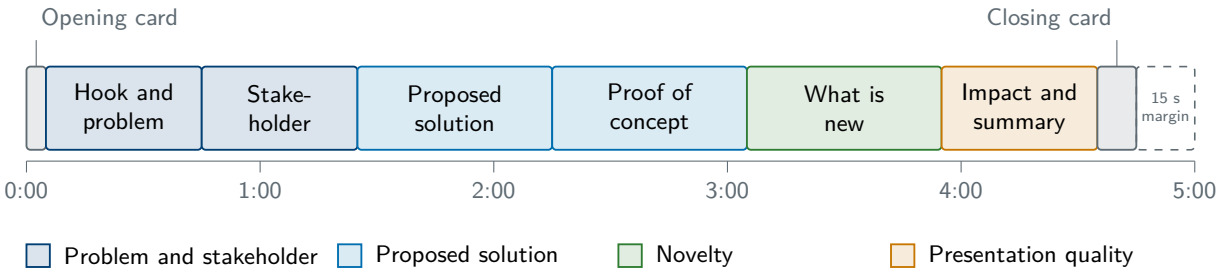


Figure 5: A suggested 5-minute running order, drawn to scale. Colours show which marking criterion each segment serves.

8.5 When things go wrong

1. Raise the problem early in a group meeting and record it in the minutes. Most problems come from unclear tasks or deadlines.
2. If it is not resolved within a week, any member emails the TA with the minutes. Do this before Monday 7 December 2026 where you can, so there is time to act before the Part B deadline.
3. The module team meets the group and agrees an action plan.
4. If a member still does not contribute after a written warning, the module team may lower that member's Part B mark through peer moderation (Section 10.3) or set them an alternative individual task.

If illness or other circumstances affect you, tell your group and apply through the College's extension or mitigating circumstances process. One member's circumstances do not normally move the group deadline, but approved circumstances are taken into account in peer moderation.

8.6 Peer assessment

- After the Part B deadline, use the peer-assessment form on Canvas to rate every member of your group, including yourself, from 1 to 5 on five criteria:
  (a) attendance and communication;
  (b) technical contribution (data, code, experiments, analysis);
  (c) contribution to the video;
  (d) contribution to the research, stakeholder and novelty analysis;
  (e) reliability: delivering agreed work on time.
- Ratings are confidential. Nobody in your group sees who gave what.
- Any rating of 1 or 2 needs a short reason with evidence, such as minutes or messages.
- The form closes at **23:59 on Wednesday 23 December 2026**. Every student must complete it; the module team follows up with anyone who has not.
- Section 10.3 explains how the ratings adjust Part B marks.

9 How to submit

Everything is submitted through **Canvas**. Part A is submitted by you, as one zip, and you share your notebook with the TA. Part B is submitted once by your group's coordinator, as three files plus an email, and every member then completes the peer assessment.

9.1 Part A, step by step

Deadline: Sunday 13 December 2026, 23:59 UK time.

1. **Check the notebook.** Restart and run all: no errors, outputs saved, code commented (Section 5).
2. **Build your model folder**, named with your CID, and run the self-check in a fresh Colab runtime (Section 5.3).
3. **Export your report** as `DL_Coursework_YourName_CID.pdf`.
4. **Build the zip** `DL_Coursework_YourName_CID.zip` with the layout below.
5. **Unzip it in a new place** and run the self-check on the extracted model folder.
6. **Upload the zip to the Part A assignment on Canvas** before the deadline.
7. **Share your notebook** with the TA, Lan Wei (`l.wei24@imperial.ac.uk`). In Colab: *Share*, add the address, *Viewer* access. A local Jupyter notebook is covered by the copy in `code/`. Do not edit the shared notebook after the deadline: the copy in your zip is the one that is marked.

```
DL_Coursework_YourName_CID.zip
|-- DL_Coursework_YourName_CID.pdf   # report
|-- code/                            # notebooks / scripts: preprocessing, training,
|   `-- ...                          # evaluation, tuning
`-- 01234567/                        # model folder, named with your CID
    |-- predict.py
    |-- shape_model.pt, force_model.pt
    |-- *.py                         # optional helper modules
    `-- requirements.txt             # optional
```

Building the zip in Colab. Download your notebook with *File* → *Download* → *Download .ipynb* and upload it back into the session's `code/` folder, or build the zip on your own computer.

```
import shutil
from google.colab import files

# 'submission' holds the report PDF, code/ and your CID model folder
name = "DL_Coursework_JaneSmith_01234567"
shutil.make_archive(name, "zip", "submission")

import zipfile
print(zipfile.ZipFile(name + ".zip").namelist())   # check before download
files.download(name + ".zip")
```

9.2 Part B: what the coordinator submits

Deadline: Sunday 20 December 2026, 23:59 UK time. The **coordinator** uploads the group files once, for the whole group, to the Part B group assignment on Canvas. `XX` is your two-digit group number.

Item	By	Where	File name

Video	Coordinator	Canvas group assignment	`GroupXX_TactileDL_Video.mp4`, plus `GroupXX_TactileDL_Video.srt` if captions are not burned in
Proof-of-concept code	Coordinator	Canvas group assignment	`GroupXX_TactileDL_Code.zip`
Contributions and AI-use form	Coordinator	Canvas group assignment	`GroupXX_TactileDL_Contributions.pdf`
Confirmation email	Coordinator	The TA, copying all members	See below
Peer assessment	Every student	Peer-assessment form on Canvas	–

If the video is too large for Canvas, upload it to your College OneDrive, give the module team view access, and submit a text file with the link on Canvas before the deadline. Do not change the file after the deadline.

The code zip. The proof-of-concept code is not graded, but it must run. One Colab notebook should run top to bottom in a fresh session; put a switch `TRAIN = False` near the top so it can load your saved weights instead of retraining; fix seeds and print library versions. Do **not** include the dataset.

```
GroupXX_TactileDL_Code.zip
|-- GroupXX_TactileDL.ipynb
|-- README.md                 (dataset, subset used, how to run)
|-- requirements.txt          (optional)
`-- models/
    `-- best_model.pt         (or .keras; over 100 MB: link in README instead)
```

Saving and zipping in Colab (copy the notebook and README into the `submission` folder first):

```
import os, shutil
from google.colab import files

os.makedirs('submission/models', exist_ok=True)
torch.save(model.state_dict(), 'submission/models/best_model.pt')   # PyTorch
# model.save('submission/models/best_model.keras')                  # Keras

# Creates GroupXX_TactileDL_Code.zip from the contents of 'submission'
shutil.make_archive('GroupXX_TactileDL_Code', 'zip', 'submission')
files.download('GroupXX_TactileDL_Code.zip')
```

The contributions and AI-use form. One PDF, agreed by the whole group, containing the contribution table (who did what, with evidence and agreed shares of effort adding up to 100%) and the group's AI-use declaration. A template is in Appendix A.6.

Part B confirmation email. The coordinator sends one email to the TA, copying every member, and gives viewer access to the Colab notebook. The TA's address is `l.wei24@imperial.ac.uk`.

```
Subject: [DeepLearning] Final Assessment PartB Submission - GroupXX

Group number:
Project title:
Project direction (A-G) and the paper(s) it builds on:
Members (name, College email):
```

```
Colab notebook link (viewer access):
Video link (only if not uploaded to Canvas):
Model weights link (only if a file is over 100 MB):
Anything the markers should know:
```

9.3 Early bonus, late penalty and extensions

Submitting early earns up to 5 bonus marks. Each day late costs 5 marks, and after 7 days late the item scores zero. Figure 6 shows the windows around a deadline.

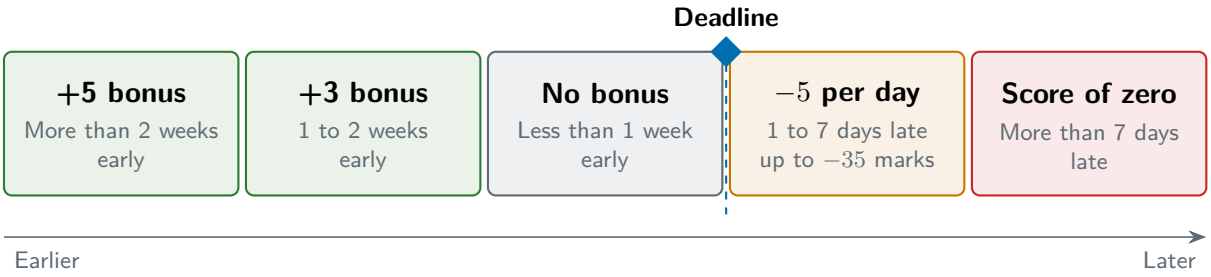


Figure 6: Early bonus and late penalty windows around a deadline. Bands are not to scale. On-time submissions in the final week earn no bonus and lose nothing.

Part	+5: more than 2 weeks early	+3: at least 1 week early
Part A	Submitted before 23:59, Sunday 29 November 2026	Submitted by 23:59, Sunday 6 December 2026
Part B	Submitted before 23:59, Sunday 6 December 2026	Submitted by 23:59, Sunday 13 December 2026

- **Early bonus.** For Part B, every group file must be in by the cut-off, the bonus goes to every member, and replacing a file afterwards removes it. You can earn an early bonus for each part, but all bonus marks together are capped at 15.
- **Late penalty. 5 marks are deducted for each day late**, for up to 7 days. After 7 days late, the item receives **zero**. Late Part B files cost the whole group.
- **Extensions** may be granted in exceptional circumstances only. Request one **in advance**, with appropriate documentation, through the College process. The coordinator asks on the group's behalf when a problem stops the whole group from submitting. Where College policy differs from this handbook, College policy applies.
- Early bonus marks count towards the 15-mark bonus cap (Section 10.4).

Worked examples: submitting Part A on 3 December earns +3. Submitting Part B 3 days late costs every member 15 marks.

9.4 Technical problems

- Upload a day early. Large videos can take a long time to upload.
- If Canvas fails near the deadline, email the TA *before* the deadline with a share link to the files and a screenshot of the error. This counts as on time if the files are not changed afterwards.
- Only the coordinator uploads Part B files. The last upload before the deadline is marked.

- Run the checklists in Section 14 before uploading.

10 How your project is marked

Part A is marked out of 60 and Part B out of 40. Your Part B mark is your group's mark multiplied by your peer factor. Bonus marks are then added, up to 15, and the final mark is capped at 100 (Equation (1)). The tables below follow the official marking criteria; the last column points to the guidance in this handbook.

10.1 Part A: individual technical coursework (60 marks)

Task	Criterion	Marks	What markers look for	Guide
1. Data preparation	Data loading and structure understanding	4	Correct scripts to load images and parse `labels.csv` for both tasks; images correctly associated with shape labels and force labels, including press/shear pairs	3.3
1. Data preparation	Exploratory data analysis	3	Sample images visualized; class and force distributions analysed; key patterns identified, such as near-zero tangential force in normal-press images	3.3
1. Data preparation	Normalization and splitting	3	Pixel intensities normalized and force targets standardized correctly; 60/20/20 split, stratified by class for shapes and grouped by `contact_id` for forces	3.3
1. Data preparation	Data augmentation (optional)	+2 bonus	Effective augmentation (rotation, flipping, colour jitter, etc.), justified for tactile sensing, e.g. flips change the sign of f_x, f_y	3.3
2. Shape classification	Model design and implementation	10	At least three CNN architectures tailored for 10-class shape classification; experiments that balance accuracy and computational efficiency	3.4
2. Shape classification	Optimization and regularization (optional)	up to +4	Techniques applied, with evidence of their effect	3.4
2. Shape classification	Evaluation metrics	5	Accuracy, Precision, Recall, F1 score and Confusion Matrix; clear interpretation of results, and comparisons	3.4
2. Shape classification	Discussion and analysis	5	Insight into model performance, failure cases (e.g. confused shape pairs) and improvements	3.4
3. Force estimation	Model design and implementation	10	A regression model for 3D contact force, with continuous outputs for f_x, f_y (tangential) and f_z (normal)	3.5
3. Force estimation	Optimization and regularization (optional)	up to +4	Techniques applied, with evidence of their effect	3.5
3. Force estimation	Evaluation metrics	5	RMSE per axis and overall; tangential error analysis for press vs. shear images against a mean-prediction baseline; comparison across architectures	3.5
3. Force estimation	Discussion and analysis	5	Insight into tangential vs. normal force errors, the role of marker displacement, and failure cases	3.5

Task	Criterion	Marks	What markers look for	Guide
4. Hyperparameter tuning	Systematic tuning (optional)	up to +5	A systematic search, with chosen values justified by experiments	3.6
5. Report	Report writing and visualization quality	10	Follows the report template; clear documentation and presentation of methods and results; quality of figures and tables	4

Required marks: 10 (Task 1) + 20 (Task 2) + 20 (Task 3) + 10 (Task 5) = 60. Hidden-test results from your `predict.py` are used as evidence of performance and generalization when marking Tasks 2 and 3 (Section 5.3).

10.2 Part B: group project video on advanced tactile perception (40 marks)

Criterion	Sub-criterion	Marks	What markers look for	Guide
Problem and stakeholder	Clarity and relevance	5	The tactile perception problem is clearly defined, and its relevance to the stakeholder, including why touch helps, is effectively argued	7.3
Problem and stakeholder	Stakeholder analysis	5	The stakeholder's needs, objectives and potential impact are thoroughly understood and concisely presented	7.3
Proposed solution overview	Feasibility analysis	5	The methods, models, datasets and metrics are realistic and suited to the problem; risks are recognized; the proof of concept supports the plan	6.6
Proposed solution overview	Creativity and innovation	5	Novelty in the methods: architecture, learning strategy, sensing combination or evaluation. Novelty in the dataset is not the key	7.3
Novelty	Open	10	Originality of the overall idea, framing or insight, clearly positioned against the papers it builds on and the obvious approach	7.3
Presentation quality	Clarity and structure	5	Well organized, flows logically, and keeps to the 5-minute limit	7.3
Presentation quality	Engagement and delivery	5	Engaging, confident and professional, with clear and concise communication; every member presents, and the video works as one piece	7.3

A proof-of-concept model is mandatory. Its code is submitted as supplementary material and is not graded.

10.3 Peer moderation of Part B

Every member normally receives the group's Part B mark. Peer assessment (Section 8.6) adjusts it only where contributions were clearly unequal.

1. Your **peer score** is the average of the totals you receive from every member, including yourself (5 criteria × 1–5, so 5–25).
2. Your **peer factor** is PF = your peer score/your group's average peer score.

3. If $0.90 \leq PF \leq 1.10$, it is set to 1: you receive the group mark.
4. Otherwise PF is used as calculated, kept between 0.50 and 1.10. Your Part B mark $B \times PF$ never exceeds 40.
5. The module team checks every adjustment against the charter, minutes, contribution logs and contributions form before applying it, and can override ratings that look unfair or coordinated.
6. Where non-contribution is documented through the process in Section 8, PF can go below 0.50, down to zero.

Worked example: group Part B mark 32, group average peer score 20

Peer score 19	$PF = 0.95$, inside the band, set to 1	Part B mark 32
Peer score 15	$PF = 0.75$	Part B mark 24
Peer score 23	$PF = 1.15$, kept at 1.10	Part B mark 35.2

10.4 Bonus marks

Source	Up to	Condition
Data augmentation (Part A, Task 1)	2	Effective and physically justified augmentation
Optimization and regularization, shape (Part A, Task 2)	4	Applied and evidenced
Optimization and regularization, force (Part A, Task 3)	4	Applied and evidenced
Hyperparameter tuning (Part A, Task 4)	5	Systematic and justified
Novel approach (Part A)	5	A creative, innovative approach, well documented and justified (Section 3.8)
Early submission	3 or 5	1 week early earns 3; more than 2 weeks early earns 5 (Section 9.3)

- No student can receive more than **15 bonus marks** in total, even though the maxima above add up to more.
- The final mark, including bonus marks, **cannot exceed 100**.
- Bonus marks are awarded against these criteria **at the instructors' discretion**.
- Focus on quality and creativity rather than collecting bonus marks: one well-justified extension is worth more than three rushed ones.

10.5 Grade descriptors

These bands apply to every criterion above. Half marks are allowed: for example, a 10-mark criterion judged "very good" scores 6 or 6.5.

Band	Share of the marks	Description
Excellent	70–100%	Complete, correct and insightful; goes beyond the minimum with clear reasons; professional presentation

Very good	60–69%	Complete and correct with minor gaps; choices mostly justified
Good	50–59%	Main requirements met; some errors, or thin justification or analysis
Adequate	40–49%	Requirements partly met; notable errors or omissions
Insufficient	0–39%	Missing, incorrect or not credible

10.6 From competent to excellent in Part A

	A competent report...	**An excellent report also...**
Data	loads the data correctly, plots class and force distributions and uses the required splits	finds non-obvious structure in the data, checks for leakage, and lets what it found shape the method
Models	trains the required number of standard architectures	chooses architectures to test a hypothesis about tactile images, and controls for confounds (training budget, resolution, pretraining)
Evaluation	reports the required metrics on the test split	reports variation across runs, includes baselines, and links numbers to specific physical causes
Tangential analysis	fills in the press/shear table and plots	explains *why* shear is harder from the sensor physics, and tests that explanation with an experiment
Background	cites the dataset paper and standard deep learning papers	positions the work in the tactile sensing literature and compares critically with published approaches
Writing	is clear and follows the template	is concise, every figure earns its space, and limitations are stated honestly

Reading the criteria strategically

- **Analysis is worth as much as results.** Discussion and analysis is worth 10 marks across Part A Tasks 2 and 3, and the tangential analysis is required.
- **Evidence must be visible.** Markers credit what they can see in your report, code and video. A good experiment that is not shown is hard to credit.
- **AI tools can write a training loop; they cannot see your experiments.** Marks go to observations from your own data, results and failure cases from your own runs, and reasons for your own design choices.
- **In Part B, novelty carries 15 of the 40 marks** (Novelty, and Creativity and innovation). Make the case for what is new explicitly, against the papers you build on.

11 Academic integrity, AI tools and data

Everything you submit must be your own work (Part A) or your group's own work (Part B), with every source cited. You may use AI tools within the limits below, provided you declare how you used them.

11.1 Academic integrity

- **Part A is individual.** Do not share Part A code, models, results or reports with anyone, including members of your Part B group.
- **Part B is shared work.** Every member is responsible for the integrity of the whole group submission, so read and watch all of it before it goes in.
- **Cite everything you did not create:** papers, datasets, pretrained weights, adapted code, figures, images and music. Code from tutorials, GitHub or online help sites is fine if you cite it in a comment and the README, and you understand it.
- **Discussing ideas with other groups is fine.** Sharing code, results or files with them is not.
- **Never fabricate or adjust results.** Report the numbers your code produces, including what did not work. Never choose models or settings on the test set and then report those numbers as test results.
- Suspected misconduct is handled under the College's academic misconduct procedures. The module team may ask any student to explain any part of a submission in a short conversation.

11.2 Using AI tools

Use	Allowed?	Conditions
Brainstorming, explaining concepts, debugging help	Yes	Declare it
Generating or completing code	Yes	Declare it; you must understand, test and be able to explain every line
Improving the grammar and clarity of text you wrote	Yes	Declare it
Help with video editing, captions or graphics	Yes	Declare it; label AI-generated images in the video
Writing whole sections of your Part A report	No	The ideas, analysis and wording must be yours
AI voices, avatars or deepfakes presenting instead of members	No	See Section 7
Generating or changing data, results or references	Never	This is fabrication. Check that every reference exists and says what you claim

Declare your use in the same four-line format: **Part A** after the references of your report; **Part B** in the contributions and AI-use form (Appendix A.6). If you used no AI tools, say so.

```
AI use declaration
Tools used (name and version):
What we used them for:
Which parts of the submission they affected:
How we checked the output:
```

For wider guidance, see the College's AI Education Hub.

11.3 Data, safety and ethical approval

- **Part A data.** Train and validate only on the released coursework data. Do not use any other TacVerse data, or any other tactile dataset, in Part A: the hidden test set comes from the same sensors. Pretrained weights from general-purpose datasets such as ImageNet are allowed; say which ones you used.
- **Part B data.** Do not use the Part A coursework dataset, or the GelSight subsets of TacVerse, in Part B.
- **Public datasets.** Follow each licence (some are non-commercial only), cite them, and never redistribute them.
- **Collecting your own data (Part B).** Agree it with the module team first. Touching objects normally needs no ethical approval, but anything involving people (touching skin, filming people, user studies) needs College ethical approval before you start.
- **Lab equipment.** Use robots or sensors only after a safety induction and with the TA's agreement.
- **Storage.** Keep any data you collect in College storage (OneDrive or Teams), not personal accounts.

12 Suggested timeline

Run Part A and Part B in parallel. Part A comes first: aim to freeze your Part A models by the end of Week 4, so Week 5 is for the report and the self-check. For Part B, read the papers while your group forms, aim for a working proof of concept by the progress check in Week 4, and keep Week 6 for recording and editing. Figure 7 shows a 6-week plan, with Week 1 starting on Monday 9 November 2026.

Part A is due at the end of Week 5 and Part B at the end of Week 6, so the two deadlines are a week apart. Do not leave Part B until Part A is submitted: a group that starts its proof of concept in Week 6 will not finish the video.

13 Common pitfalls and how to avoid them

Most lost marks come from a few avoidable mistakes: data leakage, missing metrics or interpretation, model folders the grader cannot run, and videos that overrun.

Area	Pitfall	Why it costs marks	How to avoid it
A: data	Splitting force images by image, not by `contact_id`	Press and shear images of one contact leak across splits and inflate results	`GroupShuffleSplit` on `contact_id`; assert no overlap
A: data	Normalization or force statistics from the whole dataset	Test information leaks into training	Fit on the training set only; reuse the values for validation, test and `predict.py`
A: data	An unstratified shape split	The criteria require a stratified split	Stratify by class; report counts per split
A: data	Flips or rotations without changing f_x, f_y	The labels no longer match the images	Transform the labels, or avoid these augmentations for forces; justify the mapping
A: models	Choosing models or settings on the test set	Your test results no longer show true performance	Select on validation; use test once, at the end

Area	Pitfall	Why it costs marks	How to avoid it
A: models	Three almost identical architectures	The comparison teaches nothing and earns little	Compare meaningfully different designs, and add an efficiency or robustness axis (Section 3.4)
A: evaluation	Reporting accuracy only, or micro-averaged scores	The criteria require Precision, Recall, F1 and a Confusion Matrix	Report all of them, macro-averaged
A: evaluation	Overall RMSE only	Half the images have $f_x \approx f_y \approx 0$; the tangential analysis is required	Press vs. shear table with the mean baseline, and the shear scatter plots
A: evaluation	RMSE in standardized units	Results cannot be compared	Convert back to newtons
A: evaluation	Numbers with no interpretation	Discussion and analysis is worth 10 marks	Explain what each result shows, and why errors happen
A: model folder	Forces returned standardized, or with the f_z sign flipped	Hidden-test errors are meaningless	Undo the standardization; keep the sign convention of `labels.csv`
A: model folder	Different resizing or normalization than in training; no `model.eval()`	Predictions change silently	Share one preprocessing function between training and prediction
A: model folder	Absolute paths, downloads at prediction time, GPU-only code, missing files	The grader cannot run your folder	Relative paths, weights in the folder, `map_location`; self-check in a fresh runtime
A: report	Over 6 pages, a changed template layout, or unreadable figures	Format and visualization quality are marked	Stay within 3–6 pages; large fonts, labelled axes with units
B: scope	A project that repeats the Part A tasks	Part B must be advanced, and may not use the Part A dataset	Pick a direction from Section 6.4 and say what makes it advanced
B: data	Splitting tactile data by frame	Near-identical frames leak into the test set and inflate results	Split by object, session, trial or sensor
B: research	No link to the reading list in Table 12	Part B must build on at least one paper from it	Name the paper(s) you build on and what you add
B: solution	No proof of concept, or a result with no reference point	The proof of concept is mandatory, and a bare number is hard to judge	Compare with a trivial baseline or simpler model
B: novelty	Claiming novelty without comparison, or relying on a new dataset	Novelty is 10 marks and dataset novelty is not the key	Position the idea against the papers and the obvious approach; show evidence
B: video	A video longer than 5:00	Markers stop at 5:00; your conclusion usually falls after it	Script and time it; aim for 4:30–4:50
B: video	A member who does not present, or six disconnected talks	Delivery requires every member and one coherent video	Plan hand-overs and one slide template from the storyboard
B: video	Uncredited images, footage or music	Rights and credits are required	Credit everything in the closing card
B: group	Problems raised only in the peer assessment	Moderation relies on minutes and logs	Use the process in Section 8 as soon as a problem appears
Both	Leaving it to the last hour	5 marks are lost per day late, and uploads can fail	Aim to submit early; there is a bonus for it

	Milestones	Part A individual	Part B data and model	Part B video
Week 1 9–15 Nov	Released Tue 10 Nov Start forming groups	Load data, parse labels Exploratory analysis 60/20/20 splits	Read the Part B reading list	
Week 2 16–22 Nov	Register your group on Canvas	Run the example Three shape CNNs	Shortlist directions and datasets	
Week 3 23–29 Nov	23 Nov: groups confirmed 29 Nov: char- ter, direction	Force regression Mean baseline	Download data Leakage-free split	Stakeholder and problem statement
Week 4 30 Nov–6 Dec	Progress check with the TA	**Tangential analysis** **Freeze models** **`predict.py` self-check**	**Proof-of-concept** **model trained**	Novelty argument Storyline agreed
Week 5 7–13 Dec	**Part A due** Sun 13 Dec, 23:59	**Write the report** **Zip, upload, share**	Reference point Failure cases, demo	Storyboard and script
Week 6 14–20 Dec	**Part B due** Sun 20 Dec, 23:59		Clean notebook README and form Upload	**Record, edit, captions** **Group review**
After	Peer assessment closes Wed 23 Dec, 23:59			

Figure 7: A suggested 6-week plan. Blue boxes are fixed milestones; bold boxes are the points everything else depends on.

14 Checklists before you submit

Work through every item. If one is unticked, go back to the section it comes from.

14.1 Part A (individual, Sunday 13 December 2026, 23:59)

Data, models and results

- ☐ Every image is matched to its label through `labels.csv`; press/shear pairs are linked by `contact_id`
- ☐ EDA plots are included and interpreted: sample images, class counts, force distributions for press and shear
- ☐ Pixels are normalized and forces standardized using training-set statistics only
- ☐ Shapes split 60/20/20 stratified by class; forces split 60/20/20 grouped by `contact_id`, with no overlap
- ☐ At least three CNN architectures are compared for shapes, on accuracy and efficiency
- ☐ Shape results include Accuracy, macro Precision, Recall and F1, and a Confusion Matrix
- ☐ Misclassified shapes are analysed for patterns
- ☐ The force model outputs f_x, f_y, f_z; at least two architectures are compared
- ☐ RMSE per axis and overall is reported in newtons
- ☐ The tangential table (press vs. shear, models vs. mean baseline), the shear scatter plots and the worst tangential predictions are included and discussed
- ☐ The test splits were used only for the final evaluation

Model folder and submission

- ☐ The model folder is named with your CID, is under 200 MB and contains `predict.py`, the weights and any helper modules
- ☐ `predict.py` passes the self-check in a fresh Colab runtime for both tasks, on CPU and GPU
- ☐ Shape output: `filename, class_id`; force output: `filename, fx, fy, fz` in newtons with the original sign convention
- ☐ No absolute paths and no downloads at prediction time; any extra packages are pinned in `requirements.txt`
- ☐ The notebook is commented, runs without errors and has its outputs saved; `code/` reproduces the reported numbers
- ☐ The zip is named `DL_Coursework_YourName_CID.zip` and contains the report PDF, `code/` and the model folder
- ☐ The zip is uploaded to the Part A assignment on Canvas before the deadline
- ☐ The notebook is shared with `l.wei24@imperial.ac.uk`

Report

- ☐ Uses the IEEE conference template, double column, 3–6 pages, layout unchanged
- ☐ Has all required sections: Abstract, Introduction, Objectives, Methodology, Experiments, Evaluation, Conclusions and Future Work, plus References
- ☐ No symbols, maths or footnotes in the title or abstract; abbreviations defined at first use
- ☐ Every figure and table is cited in the text, readable at column width, with labelled axes and units
- ☐ Every source, dataset, pretrained model and adapted piece of code is cited
- ☐ Any bonus work (augmentation, regularization, tuning, novel approach) is documented and justified
- ☐ The AI-use declaration follows the references

14.2 Part B (group, Sunday 20 December 2026, 23:59)

Proof of concept

- ☐ One advanced task and one dataset, with the subset documented; not the Part A dataset
- ☐ Split by object, session, trial or sensor, never by frame; normalization from the training set only
- ☐ A trained model compared with a reference point on the test set, with the right metrics
- ☐ At least one failure case explained, and a demo clip recorded
- ☐ The notebook runs top to bottom with `TRAIN = False`; README included; no raw data in the zip

Video

- ☐ 5:00 or shorter, including the opening and closing cards
- ☐ A specific stakeholder and a clearly defined tactile problem, with why touch helps
- ☐ The reading-list paper(s) that the idea builds on are named and cited on screen
- ☐ The proof-of-concept result and demo are shown
- ☐ What is novel is argued against the papers and the obvious approach

- ☐ The main risks and operating limits are named, with practical mitigations
- ☐ Every member presents in their own voice and is named on screen
- ☐ Captions burned in, or a checked `.srt` file attached
- ☐ Every third-party image, clip and music track credited

Group admin and submission

- ☐ Every file is named with your group number (`GroupXX_...`)
- ☐ The contributions and AI-use form is agreed by every member
- ☐ The coordinator has uploaded the video, code zip and form to Canvas, and sent the confirmation email
- ☐ The Colab notebook is shared with viewer access
- ☐ Every member completes the peer assessment on Canvas by 23:59 on Wednesday 23 December 2026

15 Frequently asked questions

15.1 Part A

Do I have to do both shape classification and force estimation?

Yes. Task 2 (shape) and Task 3 (force) are both required, and each is worth 20 marks. The tangential force analysis in Task 3 is also required.

Do I need three architectures for force estimation as well?

Shape classification needs at least three CNN architectures. Force estimation needs a comparison across architectures: at least two.

Can I use PyTorch or TensorFlow?

Either, or any framework pre-installed in Colab. The example submission uses PyTorch. What matters is that `predict.py` follows the interface and runs in a fresh Colab runtime.

Can I use pretrained networks or a vision transformer?

Yes. Cite them, explain any input changes, and ship the fine-tuned weights in your model folder: nothing may be downloaded at prediction time. A vision transformer can be an extra comparison; at least three of your shape models must be CNNs.

Can I use the press/shear pairing during training?

Yes: any training signal from the released data is allowed. At prediction time your model receives a single image only.

Can I retrain my final model on training plus validation data?

Yes, if you say so in the report. All reported numbers must still come from your untouched test splits.

Are data augmentation and hyperparameter tuning required?

No. Augmentation earns up to 2 bonus marks and systematic tuning up to 5. Optimization and regularization earn up to 4 bonus marks for each of the two tasks.

What data will my model be tested on?

A hidden test set held by the module team, recorded with the same sensors and preprocessed in the same way. It is never released. That is why `predict.py` must follow the interface exactly.

Is there a minimum accuracy or RMSE?
No. We mark the quality of your method, evidence and analysis. Your model folder must run, though.

Do references count toward the 6 pages?
No. References and the AI-use declaration may go beyond page 6.

My results differ from the TacVerse paper.
That is expected: the protocols differ. Explain the differences rather than chasing a number.

15.2 Part B

What counts as "advanced"?
Going beyond the single-image classification and regression of Part A: a different kind of model, a harder setting, another modality, or a new tactile property (Section 6.1).

Do we have to build the sensors in the papers?
No. The reading list is the starting point for your idea. Your proof of concept can use a public dataset (Section 6.5) or data you collect.

Can we use the Part A dataset?
No. Part A is individual work, and Part B must not reuse its data.

Can we use a dataset that is not on the list?
Yes. Name it in your direction paragraph by Sunday 29 November so the module team can check it is suitable.

Is there a minimum accuracy for the proof of concept?
No. Marks reward a feasible, well-argued and novel solution backed by an honest proof of concept, not the highest number.

TensorFlow or PyTorch for Part B?
Either. Pick one for the whole notebook.

Our video runs to 5:20. Does that matter?
Yes. Markers stop at 5:00, so whatever comes after, usually your conclusion, earns nothing, and keeping to time is part of Clarity and structure. Cut it.

With six of us, how can everyone present in five minutes?
Plan 40–50 seconds per speaker and use the running order in Section 7, which ends at 4:45. Short, well-rehearsed segments with clean hand-overs work better than long ones.

A member cannot appear on camera.
Narrating over slides in their own voice is fine. Talk to the module team about any other adjustment.

One member has done nothing for two weeks.
Start the process in Section 8 now. Do not wait for the peer assessment.

Who uploads, and which version counts?
Only the coordinator uploads Part B files. The last upload before the deadline is marked.

Is our Part B code marked?
No, but it is mandatory. Submit it as `GroupXX_TactileDL_Code.zip` with your video.

Can we post our video online afterwards?

After marks are released, if every member agrees and every piece of media in it allows public sharing.

15.3 Both parts

What happens if I submit late?
You lose 5 marks per day, for up to 7 days. After 7 days late, the item scores zero. Late Part B files cost the whole group. Extensions are possible only in exceptional circumstances, requested in advance with documentation.

Who do I contact?
Dr Dandan Zhang (module leader) and Ms Lan Wei (teaching assistant, `l.wei24@imperial.ac.uk`). Bring questions to the module's teaching sessions or email the TA; clarifications for everyone are announced on Canvas.

A Templates

Copy these into your own files or your group's shared drive.

A.1 Part A splits and metrics

```
import numpy as np
from sklearn.model_selection import train_test_split, GroupShuffleSplit
SEED = 42

# Shape classification: stratified 60/20/20
tr, tmp = train_test_split(shape_df, test_size=0.4,
                           stratify=shape_df.class_id, random_state=SEED)
va, te = train_test_split(tmp, test_size=0.5,
                          stratify=tmp.class_id, random_state=SEED)

# Force estimation: grouped by contact_id, 60/20/20
g1 = GroupShuffleSplit(n_splits=1, test_size=0.4, random_state=SEED)
tr_i, tmp_i = next(g1.split(force_df, groups=force_df.contact_id))
tmp = force_df.iloc[tmp_i]
g2 = GroupShuffleSplit(n_splits=1, test_size=0.5, random_state=SEED)
va_i, te_i = next(g2.split(tmp, groups=tmp.contact_id))
ftr, fva, fte = force_df.iloc[tr_i], tmp.iloc[va_i], tmp.iloc[te_i]
assert not set(ftr.contact_id) & set(fte.contact_id)

AX = ["fx", "fy", "fz"]
mu, sd = ftr[AX].mean(), ftr[AX].std()     # training statistics only

def tangential_metrics(df, pred):
    """df: test rows with fx, fy, fz, is_shear; pred: (N, 3) array in newtons."""
    err = pred - df[AX].to_numpy()
    rows = {}
    for name, mask in [("press", df.is_shear.eq(0).to_numpy()),
                       ("shear", df.is_shear.eq(1).to_numpy()),
                       ("all", np.ones(len(df), bool))]:
        e = err[mask]
        rows[name] = {"RMSE_fx": np.sqrt(np.mean(e[:, 0]**2)),
                      "RMSE_fy": np.sqrt(np.mean(e[:, 1]**2)),
                      "RMSE_fz": np.sqrt(np.mean(e[:, 2]**2)),
                      "RMSE_t": np.sqrt(np.mean(e[:, 0]**2 + e[:, 1]**2)),
```

```
                        "RMSE_all": np.sqrt(np.mean(e**2))}
    return rows

# Mean baseline: the training-split mean force for every test image
baseline = np.tile(ftr[AX].mean().to_numpy(), (len(fte), 1))
```

A.2 Part B group charter (upload by Sunday 29 November 2026)

```
# Group charter: GroupXX

Project direction (A–G) and the paper(s) it builds on:

| Name | College email | Lead role | Also helps with |
|------|---------------|-----------|-----------------|
|      |               |           |                 |

Communication channel and reply time (e.g. Teams, within 24 h on weekdays):
Weekly meeting (day, time, room or link):
How we decide (e.g. discuss, then majority vote; coordinator breaks ties):
Internal deadlines (proof of concept, storyboard, first full video cut):
If someone misses a deadline, we will:
Signed (name, date):
```

A.3 Meeting minutes (one per meeting)

```
# GroupXX meeting, [date]

Present:                      Apologies:
Progress since last meeting:
Decisions:

| Action | Owner | Due |
|--------|-------|-----|
|        |       |     |

Problems or risks:
```

A.4 Personal contribution log (one per member)

```
| Date | Task | Hours | Evidence (link) |
|------|------|-------|-----------------|
|      |      |       |                 |
```

A.5 Five-minute storyboard (agree before recording)

Shot	Time	Speaker	What is said (summary)	What is shown
1	0:00–0:05	–	–	Opening card: group, title, names
2	0:05–0:45		Why touch matters for our stakeholder	Raw tactile sensor footage

3	0:45–1:25		Who the stakeholder is and what they need	
4	1:25–2:15		Our proposed solution, its feasibility and key risks	Architecture diagram
5	2:15–3:05		Proof-of-concept result and demo	Results table, demo clip, a failure case
6	3:05–3:55		What is new, against the papers we build on	Paper references, comparison with the obvious approach
7	3:55–4:35		Impact and call to action	
8	4:35–4:45	–	–	Closing card: who did what, datasets, credits

A.6 Part B contributions and AI-use form

Submit as `GroupXX_TactileDL_Contributions.pdf`. Agree it as a group. Shares of effort add up to 100%. Give evidence: notebook parts, figures, video segments, research notes.

Member	**Lead role**	**Main contributions, with evidence**	**Share (%)**
A. Name	Data lead	Data loader and split (notebook part 2), sample-image figure, video segment 3	
	Coordinator and stakeholder lead		
	Research and novelty lead		
	Modelling lead		
	Evaluation and demo lead		
	Video producer		
Total			100

```
AI use declaration (GroupXX)
Tools used (name and version):
What we used them for:
Which parts of the submission they affected:
How we checked the output:

Agreed by (every member, name and date):
```

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